# OpenAffect-EEG: Auditing EEG Added Value Under Matched Label Resources

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## Abstract

Attributing an affect-prediction gain to EEG requires a comparator with the same non-EEG label resources. A comparison against an uncalibrated baseline instead evaluates the system package. OpenAffect-EEG makes this distinction executable. It fixes test trials, independently doses participant-calibration and repeated-stimulus labels, preserves population-training size, and pairs each EEG predictor with an EEG-free predictor receiving the same labels. On EmoEEG-MC and urban-image appraisal, we evaluate complete $5 \times 5$ dose grids for two settings each of band-power regression, frozen LaBraM, and standard EEGNet: 1,500 model cells in total. At maximum dose, positive system gains of $+0.0961$ and $+0.0587$ for one LaBraM setting become matched EEG differences of $-0.0459$ $[-0.1098, +0.0139]$ and $-0.0186$ $[-0.0563, +0.0126]$, respectively. Across all 12 task–setting grids, intervals for the mean matched EEG increment include zero. These results do not establish equivalence or absence of EEG information; they show that the observed total gains cannot be attributed to EEG under the evaluated contracts. Crossed simulations verify intended positive-signal behaviour but reveal serious undercoverage on the primary five-block topology; the reported intervals are conditional diagnostics, not calibrated guarantees. A companion checker records trial lineage, predictor resources, and deployment requirements, returning allow, block, or unverifiable decisions. OpenAffect-EEG contributes a resource-matched evaluation object and auditable evidence, not a new encoder or a causal decomposition of emotion.

## 1 Introduction

An EEG affect system can use more than the current EEG recording. It may also receive ratings from the target user's earlier trials, ratings of the same stimulus from other users, or source-released normative stimulus scores. These resources can be legitimate at deployment. The attribution problem begins when an EEG system receives them but its EEG-free comparator does not. If the former scores higher, the difference combines EEG with unequal label access. Canonical affect datasets make this problem concrete because they repeatedly present stimuli and collect individual self-reports [6, 10, 30].

This paper asks one diagnostic question: *on identical test trials, does a reported gain remain when EEG and EEG-free predictors receive the same participant and stimulus label resources?* Trial-level leakage prevention is necessary but does not answer this question. A model can exclude every test trial while still being compared with a baseline that lacks its calibration labels. Conversely, resource matching does not restore an unseen-participant or unseen-stimulus claim once those identities have been exposed. The two checks address different scientific statements.

OpenAffect-EEG turns the comparison into a controlled evaluation object. It separates total system gain into an EEG-free calibration gain and a matched EEG difference, while fixing test support, controlling population-training size, and recording every supplied label resource. The subtraction is elementary; the contribution is the executable design that makes its operands comparable. This follows the diagnostic-benchmark principle that aggregate performance can hide which input or capability produced it [25], but applies that principle to deployment resources in EEG affect prediction.

The central empirical correction is concrete. In two tasks, favourable gains against an uncalibrated prior become uncertain when compared with an EEG-free predictor receiving the same calibration labels. We do not conclude that EEG is uninformative. We conclude that the favourable total gains observed here do not identify its incremental contribution. We make three contributions:

1. **A resource-matched evaluation object.** A fixed-support response surface independently doses target-participant calibration and other-participant ratings of test stimuli. Training-row replacement and paired EEG-free controls isolate the score contrast of interest without claiming causal identity or emotion disentanglement.
2. **A controlled empirical audit.** Complete dose grids for three EEG model families, two tasks, paired crossed-bootstrap uncertainty, and known-truth simulations test whether total gains survive the matched comparator and where the evaluation itself becomes unreliable.
3. **An executable claim contract.** Trial lineage, predictor resources, preprocessing and selection roles, and fixed test hashes produce allow, block, or unverifiable decisions. Pinned external paths and synthetic faults expose the difference between a valid split and a supported deployment claim.

# 2 Relation to Existing Evaluation Work

EEGain standardizes datasets, preprocessing, models, splits, metrics, and prediction logging [11]. FMScope audits frozen representations with subject-axis, spectral, layer, and within-subject diagnostics [17]. Calibration, joint person–stimulus holdout, and EEG toolboxes are also established [3, 15, 19]. These efforts ask whether data partitions, models, or representations generalize. Our unit of analysis is instead a *paired prediction contrast*: which labels each predictor received on a fixed test support, and what part of a system gain remains after matching those labels.

Identity probes, linear erasure, data cards, and provenance hashes are not claimed as individually novel [1, 8, 17, 24]. Outside EEG, diagnostic benchmarks separate aggregate scores into controlled capabilities, stress-test metrics on configurable known-truth data, and pair standard protocols with reusable code [20, 25, 26, 28]. OpenAffect-EEG adopts those benchmark qualities for a domain-specific estimand: resource-matched EEG added value in observed self-report prediction. Dose curves, simulation, and checklists are supporting mechanisms, not separate novelty claims. We neither audit physiological mechanisms nor claim broader dataset coverage or higher classification accuracy than existing EEG suites.

# 3 Resource-Matched Evaluation

## 3.1 Trial graph and resource contract

Each trial has a participant identity $i$, stimulus identity $j$, observed report $y$, context, and source lineage. Windows inherit their complete trial's partition. Let $T_b$ be the fixed test trials in identity block $b$ and $D_b^{p,s}$ the population development data at doses $(p, s)$. Let $C_{i,b}^p$ contain $p$ labelled calibration trials from target participant $i$, on stimuli outside the test stimulus set. Repeated-stimulus dose $s$ counts population ratings per test stimulus from other participants. We record a contract

$$\mathcal{R}_b(p, s) = \big(T_b, D_b^{p,s}, \{C_{i,b}^p\}_i, N_b, A_b, F_b, V_b\big), \tag{1}$$

where $N_b$ is the fixed population-training row count, $A_b$ lists additional label resources such as normative scores, and $F_b, V_b$ record preprocessing fit and model-selection trials. This notation describes actual supplied resources; it is not an identity-information measure.

Repeated-stimulus rows replace context- and target-matched training donors, so population row counts remain fixed. Calibration rows do not fit the population model; they estimate a post-fit participant mean residual offset. Dose sets are deterministic and nested. Removing donors changes training composition; we control count and approximate label/context balance, not every property of

Table 1: Complementary evaluation objects. This is a scoped comparison of published purposes, not an exhaustive absence-of-feature claim.

| Approach | Question addressed | Additional object evaluated here |
|---|---|---|
| Trial/identity holdout | Do forbidden trials or identities cross partitions? | Dose-controlled resources with a fixed test and population size. |
| EEGain [11] | How do models compare in a standardized pipeline? | Does an EEG gain persist against a comparator with the same calibration labels? |
| FMScope [17] | What identity structure is encoded, and how does erasure affect decoding? | What labelled resources supported this prediction contrast and deployment claim? |
| OpenAffect-EEG | What can be attributed to EEG under a declared resource contract? | Paired score contrasts plus verifiable resource and execution records. |

the training distribution. Nor is total labelled-data budget fixed across the participant axis: additional calibration is the resource being measured.

The principal invariants are: no test trial in fit, selection, or calibration; identical test trials across cells and models; exact realised doses; no target participant in population fitting; and identical non-EEG labels for paired predictors. Test identities are initially jointly held out, but at nonzero doses the corresponding identities have been exposed. Released normative scores, where available, remain separately declared at zero stimulus-label dose. Zero dose therefore does not generally mean no stimulus knowledge. Unverifiable session, device, or upstream pretraining identities cannot support a strict isolation certificate.

### 3.2 Predictors and the estimand

The population prior $P$ uses fit-set stimulus means, falling back to declared source normative ratings or the fit-global mean. $PC$ adds the target user's mean calibration residual without EEG. $E$ adds an EEG-predicted population residual to $P$, and $EC$ calibrates $E$ using exactly the same trial labels as $PC$. Both calibrated predictors use the same mean-offset rule. The residual is arithmetic relative to the specified prior, not an identified private emotion. Direct EEG prediction is retained as a secondary control.

For score functional $Q$ on the same $T_b$, define

$$G_{\text{total}}(p,s) = Q(EC) - Q(P), \tag{2}$$

$$G_{\text{cal}}(p,s) = Q(PC) - Q(P), \tag{3}$$

$$\Delta_{\text{EEG}}(p,s) = Q(EC) - Q(PC). \tag{4}$$

Thus $G_{\text{total}} = G_{\text{cal}} + \Delta_{\text{EEG}}$ exactly. This is a telescoping score identity, not an additive decomposition of CCC into neural mechanisms. At a fixed $s$, both $P$ and $EC$ already receive the same repeated-stimulus labels: the unmatched resource in this particular contrast is participant calibration. Changing $s$ studies the matched comparison under different shared stimulus resources, rather than estimating a pure causal effect of stimulus information. $G_{\text{total}}$ can assess an entire system package; it cannot isolate EEG added value when calibration resources differ. At $p = 0$, $PC = P$ and the matched contrast reduces to $Q(E) - Q(P)$.

Our primary summary is the uniform mean of $\Delta_{\text{EEG}}$ in CCC over five fixed identity blocks and the complete grid $\{0, 1, 2, 4, 8\}^2$. We report paired MAE differences as a secondary metric; positive MAE differences mean greater error. Uniform grid weighting is a specified benchmark utility, not an estimated deployment-frequency distribution. Endpoint tables illustrate comparison choice but do not replace this primary summary. Every grid cell is retained in the aggregate release.

This estimand is specific to the fitted predictors and calibration rule. Equal label access does not imply equal ability to exploit those labels. Accordingly, a nonpositive increment diagnoses the evaluated pipeline, not the maximum information available in EEG. The uniform mean assigns equal weight to benchmark cells; it is not an optimal deployment policy. For an application, resource feasibility, participant time, recording cost, and application-specific cell weights must be specified separately.

### 3.3 Observable execution checks

The checker consumes a trial-identity table, an execution record, and boolean deployment requirements. Unlike a split-name lookup, it compares executed UID sets for population labels, calibration labels, test support, preprocessing, and model selection. Required missing evidence yields `unverifiable`, not a pass. A detected violation yields `block`; otherwise supplied observable evidence may `allow` the requested checks. Source identity authenticity and upstream pretraining exposure remain unverifiable even when observable checks pass.

For example, a calibrated target-user system can pass test-trial isolation yet fail a requested unseen-participant contract. Two predictors can also share every train/test split but fail resource parity because only one receives calibration labels. These are different checks with different inputs. A basic intersection checker is not expected to infer undeclared resource use. The resource checker complements it; it is not a universally more accurate detector.

## 4 Controlled Evidence

### 4.1 Evaluation coverage

We evaluate EmoEEG-MC (30 eligible participants, 42 stimuli; valence and arousal) and Urban Appraisal (62 participants, 56 street images; genuine valence trials only) [30, 32]. Deterministic identity blocks test every eligible participant and stimulus once, yielding 252 and 695 fixed test trials. The design tests five diagonal blocks, not every possible participant–stimulus pair. Urban is an external appraisal task, not an identical replication of emotion induction, and neither dataset is an untouched confirmatory cohort.

Each task includes band-power Ridge, frozen LaBraM, and standard Braindecode EEGNet [9, 12]. Two bounded settings per model, five identity blocks, and all 25 dose cells yield 1,500 model-cell results; these cells are not independent observations. Ridge settings vary the candidate regularization grid, while EEGNet settings vary learning rate. Validation selects hyperparameters and training duration. Population refitting uses training and validation trials but never test or calibration trials. We record 200 nonconstant EEGNet selection/refit histories as execution evidence, not as proof of optimization. The Appendix reports full preprocessing, coverage, budgets, and historical model panels.

The archived learning records quantify this qualification. Among 50 direct and 50 residual EEGNet fits per task, respectively 12/10 Emo fits and 8/7 Urban fits select epoch one. Median selected epochs are 18/17.5 and 11/7.5. Training objectives decrease in the median for all four groups. These real-task records rule out an account in which every network simply remained at its initial fit; they do not establish optimal tuning or a positive held-out EEG signal. Appendix C reports the reusable audit.

### 4.2 A favourable total gain need not be an EEG gain

Table 2 applies the same predeclared maximum dose to every setting. For EmoEEG-MC LaBraM R1, $EC$ improves on the uncalibrated prior $P$ by $+0.0961$. Yet the same calibration labels improve the EEG-free predictor by $+0.1420$, leaving a matched EEG difference of $-0.0459$ $[-0.1098, +0.0139]$. In Urban, the corresponding quantities are $+0.0587$, $+0.0774$, and $-0.0186$ $[-0.0563, +0.0126]$. The system-level point gains are positive; the EEG-specific contrasts are not established as positive. This is the judgement that resource matching changes.

The primary full-grid summaries in Table 3 tell the same bounded story: all 12 task–setting intervals include zero under both interval methods. No equivalence margin was prespecified, so this does not establish equivalence, universal reversal, absence of EEG information, or a ceiling on future EEG methods. The secondary MAE contrast for LaBraM R1 indicates increased error in both tasks under both interval constructions; other configurations are uncertain or interval-sensitive. Appendix A reports all CCC and MAE settings rather than selected rows.

Figure 1 exposes heterogeneity hidden by the uniform mean. For example, Urban EEGNet L2 has positive point estimates in every cell, ranging from $+0.0003$ to $+0.0177$; these do not establish positive increments at the reported precision. None of the 300 cells has a positive pointwise CCC interval. Two Urban LaBraM R1 cells have negative pointwise intervals, but neither survives its within-

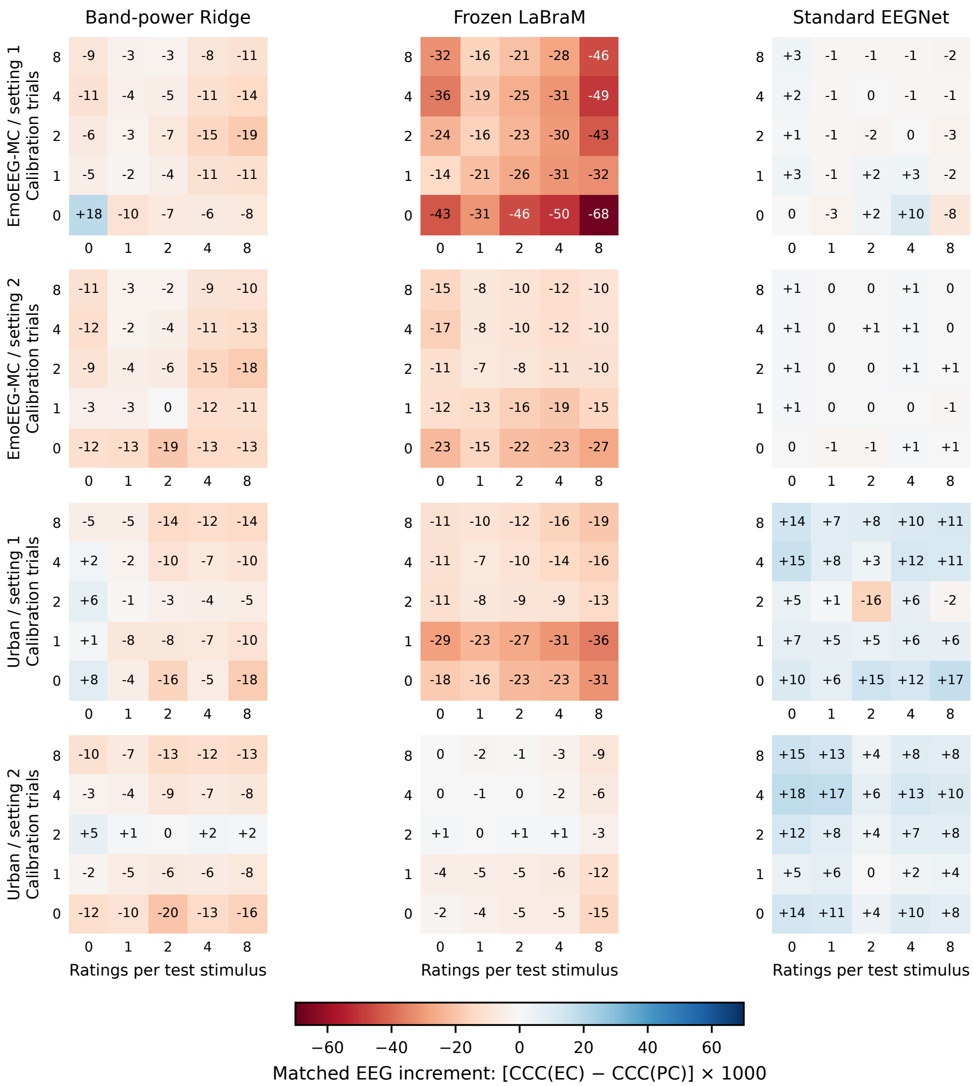


Figure 1: All 12 matched-increment grids, with every dose retained. Entries and color scale are $1000\Delta_{\mathrm{EEG}}$; rounded zero is not exact equality. Rows distinguish tasks and settings (R1/L1, R2/L2); columns distinguish model families. Positive values favor EEG. Each value averages five block-level paired CCC contrasts. This is a descriptive display of already inspected data, not an additional confirmatory test. Conditional pointwise and within-model dose-family intervals accompany every cell in the released CSV.

Table 2: Changing the comparison at the predeclared maximum dose $(p, s) = (8, 8)$. Total system gain $EC - P$ includes the EEG-free calibration gain $PC - P$. The last column is the paired percentile interval for $EC - PC$, not a difference of interval endpoints. These descriptive endpoints do not replace the primary uniform-grid mean.

| Task | Setting | $EC - P$ | $PC - P$ | $EC - PC$ | Paired interval |
|---|---|---|---|---|---|
| EmoEEG-MC | Band-power R1 | +0.1312 | +0.1420 | -0.0108 | [-0.0429, +0.0219] |
| EmoEEG-MC | Band-power R2 | +0.1318 | +0.1420 | -0.0102 | [-0.0403, +0.0206] |
| EmoEEG-MC | LaBraM R1 | +0.0961 | +0.1420 | -0.0459 | [-0.1098, +0.0139] |
| EmoEEG-MC | LaBraM R2 | +0.1325 | +0.1420 | -0.0095 | [-0.0419, +0.0243] |
| EmoEEG-MC | EEGNet L1 | +0.1400 | +0.1420 | -0.0020 | [-0.0065, +0.0027] |
| EmoEEG-MC | EEGNet L2 | +0.1420 | +0.1420 | 0.0000 | [-0.0040, +0.0043] |
| Urban | Band-power R1 | +0.0639 | +0.0774 | -0.0135 | [-0.0561, +0.0362] |
| Urban | Band-power R2 | +0.0643 | +0.0774 | -0.0131 | [-0.0398, +0.0199] |
| Urban | LaBraM R1 | +0.0587 | +0.0774 | -0.0186 | [-0.0563, +0.0126] |
| Urban | LaBraM R2 | +0.0689 | +0.0774 | -0.0085 | [-0.0373, +0.0172] |
| Urban | EEGNet L1 | +0.0885 | +0.0774 | +0.0111 | [-0.0147, +0.0360] |
| Urban | EEGNet L2 | +0.0851 | +0.0774 | +0.0078 | [-0.0064, +0.0252] |

Table 3: Primary uniform-grid mean matched EEG increment in CCC, with paired percentile intervals. Each task averages five fixed identity blocks and all 25 dose cells. Settings are sensitivity checks, not independent replications.

| Setting | EmoEEG-MC: mean [interval] | Urban: mean [interval] |
|---|---|---|
| Band-power R1 | -0.0069 [-0.0333, +0.0221] | -0.0060 [-0.0432, +0.0376] |
| Band-power R2 | -0.0091 [-0.0306, +0.0145] | -0.0069 [-0.0312, +0.0216] |
| LaBraM R1 | -0.0320 [-0.0676, +0.0104] | -0.0173 [-0.0482, +0.0112] |
| LaBraM R2 | -0.0137 [-0.0376, +0.0139] | -0.0035 [-0.0238, +0.0150] |
| EEGNet L1 | -0.0001 [-0.0040, +0.0047] | +0.0073 [-0.0056, +0.0217] |
| EEGNet L2 | +0.0003 [-0.0028, +0.0040] | +0.0086 [-0.0043, +0.0215] |

model dose-family band. We retain all 48 task–setting–corner comparisons $(0, 0), (8, 0), (0, 8), (8, 8)$ in a separate table; no cell selection or averaging of interval endpoints is used.

Why is the matched comparator material? From zero dose, eight target-user calibration trials increase EEG-free $PC$ CCC by $+0.2036$ in EmoEEG-MC and $+0.1564$ in Urban. Eight other-participant ratings per test stimulus increase it by $+0.4907$ and $+0.3408$, respectively; both interval constructions are positive. The fact that labels help is unsurprising. Their magnitude relative to the matched EEG contrasts shows why unequal label access can change the attribution attached to a system score.

### 4.3 Uncertainty and evaluator validity

The crossed bootstrap applies identical participant and stimulus multiplicities to every paired predictor, cell, and block. Of 2,000 draws, 1,985 are valid for EmoEEG-MC and 2,000 for Urban. Percentile intervals are primary; an explicitly heuristic small-cluster t-normal analysis is a sensitivity check. Both condition on fitted predictions and omit full retraining uncertainty. Pointwise intervals and separately released dose-family simultaneous bands do not correct all model comparisons.

The relevant cluster sizes are those within each scoring block: Emo has six participants and eight or nine stimuli per block; Urban has 12 or 13 participants and 11 or 12 stimuli. The whole-dataset participant count does not by itself validate a mean of small-block CCCs. We therefore distinguish the earlier single-block simulations below from the topology-matched five-block check in Appendix C.

The new topology-matched validation executes 300 replicates for each of four Gaussian scenarios per task, with 2,000 bootstrap draws and no failed runs. For trial signal, percentile coverage of the uniform mean is 0.900 for Emo and 0.927 for Urban; for participant-constant signal it is 0.903 and 0.970. The worst cell is Emo's zero-dose participant-constant case: coverage is 0.193 (Monte Carlo interval $[0.153, 0.242]$). Thus nominal 95% coverage is not established, even on the intended topology. The t-normal sensitivity improves these mean-level checks but still undercovers that cell (0.887). We retain both predeclared constructions as conditional diagnostics and make no calibrated significance or equivalence claim. Earlier single-block tests in Appendix A.5 cannot override this failure.

### 4.4 Scope checks and retained negative evidence

Earlier EmoEEG-MC and MusicEEG surfaces, compact versus standard EEGNet, CBraMod, LaBraM random controls, frozen EMOD, and calibration variants remain in Appendix F.2. They use distinct supports or protocols and are not replications of the primary grid. The earlier MusicEEG LaBraM mean increment is $0.0182$ $[-0.0161, 0.0452]$, an uncertain auditory result that we retain rather than omit. Resource-conditioned model selection likewise has no stable mean CCC benefit (Appendix F.1).

Identity probes and subspace perturbations are representation diagnostics, not causal evidence. The ds006866 analysis concerns condition exposure because image identity is unavailable. DENS is missingness-limited; SEED-V is an authorized aggregate supplement, not an openly redistributed benchmark. The repaired EMOD single-seed check is neither a full original-task reproduction nor evidence against EMOD, and mdJPT has no completed numerical audit here. These records bound model and domain coverage; none is promoted to primary evidence.

## 5 Executable Resource and Claim Audit

Table 4: Executed external paths and reusable checks. The audit concerns the specified code configuration and claim, not the validity of an original paper's reported model scores.

| Record | Observable finding | Consequence for the requested claim |
| --- | --- | --- |
| LibEER empty-validation path | The exercised branch assigns test arrays to validation. | Blocks validation-based selection claimed to exclude test trials. |
| EEGain window validation split | Generated windows of a real trial can enter both training and validation. | Blocks an additional validation-trial holdout; the same trace still passes test-participant holdout. |
| Matched-resource toy | Only the EEG predictor receives personal calibration labels. | Blocks a same-label-resource comparison despite clean trial splits. |
| Missing-provenance toy | Preprocessing fit-trial evidence is absent. | Reports unverifiable, rather than treating absence of evidence as a pass. |

We execute functions at pinned EEGain and LibEER commits, using eligible EmoEEG-MC trial identities and three generated marker windows per trial. For LibEER we also execute the precise empty-validation branch in its EEGNet training entry point; we do not train the external classifier [11, 16]. Table 4 contains two scope-sensitive examples. An ordinary split checker detects these observed split violations too. The added evaluation object is whether the executed evidence supports the requested claim: the same trace can block a validation-trial-isolation claim while allowing a narrower held-out-test-participant claim. Resource parity further requires declared predictor inputs, which a train/test intersection cannot infer.

Twenty native or scope-qualified cases contain ten requested-rule violations and ten clean or correctly qualified cases. Both checkers give the corresponding decisions. Separately, 70 developer-designed resource fixtures contain 50 injected violations, ten clean cases, and ten missing-evidence cases. The resource checker blocks, allows, and preserves uncertainty respectively. These counts verify finite rule coverage, not a literature-wide detection rate or independent user study. We do not attribute injected faults to either toolkit, or infer which configuration produced any published result.

**Reusable interface.** The source exposes the following no-EEG workflow audit:

```
python -m openaffect_eeg.workflow_audit \
  --trials trials.tsv --execution execution.json \
  --contract contract.json --output report.json
```

The report names failed checks and counts set differences. Reproduction instructions include a synthetic example with matched, unmatched, missing provenance, and falsely claimed unseen-user cases. Separate raw-data commands run model experiments on user-provided licensed data. File hashes establish byte lineage, not the truth of a supplied UID or upstream training history. The release also contains a machine-readable rule catalog and a source-linked claim-repair record: across five EEGain folds, the same pinned split function and trial universe block a joint subject-plus-validation-trial claim, while the narrower held-out-test-participant claim is allowed. This repairs the scope of the claim; it does not repair or invalidate an external model score.

## 6 Discussion and Limitations

**The judgement corrected.** We operationalize an *attribution rule*: without resource parity, an EEG system's advantage over an uncalibrated prior is not its incremental EEG value. The experiments show that this distinction changes the numerical conclusion in the evaluated tasks. They do not establish that earlier publications used the same unmatched comparison. Nor does every subject-independent protocol become invalid: a properly scoped zero-calibration claim remains meaningful. The resource ledger makes that scope explicit.

The two evidence streams have different roles. Tables 2 and 3 change the interpretation of actual EEG predictions generated in this study. The external code cases validate scoped execution claims without reproducing an external classifier. Together they support a preventive evaluation workflow; they do not measure how often published EEG gains arose from resource mismatch.

**When is the method useful?** A researcher comparing encoders can hold test and labelled resources fixed; a system designer can report a user-calibration budget alongside performance; and a reproducer can distinguish a valid trial split from a resource-matched comparison. These are supported workflow capabilities. We have not established broad adoption, independently measured usability, improved model selection, or an economic benefit from EEG acquisition. A paired no-EEG comparator is a baseline for the stated task, not a clinically validated alternative device.

**Limits of the empirical claim.** The latest analysis is retrospective, covers selected identity pairs, and uses bounded training settings. Frozen representations and post-fit calibration do not span all fine-tuning or adaptation methods. Learned preprocessing excludes test trials, but stimulus-dose injection can change validation-side identity relations; we do not claim every validation identity stays isolated. The participant and stimulus dose axes are resource interventions on a trial graph, not randomized interventions on latent affect. Extra normative labels and unverifiable pretraining exposure must remain visible in the contract. Uniform-grid averages can obscure heterogeneous cells, hence the complete release; intervals condition on fixed fits and have small-cluster limitations. The independently collected tasks differ in elicitation and target definition, so they do not establish universal EEG-emotion behaviour.

## 7 Artifacts, Ethics, and Conclusion

Code, aggregate results, split rules, source versions, model/data cards, synthetic examples, and evidence hashes accompany the work. The anonymous code entry provides unauthenticated access to the README, installation configuration, acceptance script, and workflow checker, verified on 9 September 2026. The manuscript and artifact version must be matched at submission; this file access check is distinct from full experimental reproduction. Developer-run clean-environment checks and author-reported colleague execution are recorded separately; no independent reproduction of all EEG experiments is claimed. EEG recordings, derived individual features, trial-level private predictions, stimulus materials, and model weights are not redistributed. Original consent and licensing apply; Urban's snapshot contains inconsistent license wording, documented without redistributing its recordings. Identity probes must not be used to identify participants, and self-report prediction is not access to a clinical state or private emotion.

OpenAffect-EEG provides a fixed-support, resource-matched comparison and an executable check of its observable assumptions. Its main finding concerns what a score contrast warrants: total system improvement and EEG added value must be reported separately. Uncertain matched increments are retained as uncertain; the contribution is improved evaluation specificity and reproducibility, not a requirement that EEG either succeed or fail.

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# A Latest Resource-Matched Coverage Analysis

This section and the main resource-matched tables use the frozen `final_closure_v9` evidence. The historical supplement preserves the original experiments, figures, and their support. They are not additional independent replications of this panel, and their historical use of "primary" denotes their own frozen analysis plan rather than replacing the present main summary.

## A.1 Source and coverage details

EmoEEG-MC uses snapshot 1.0.7; Urban Appraisal uses snapshot 1.0.0 [30, 32]. Emo has 1,260 eligible trials from 30 participants and 42 stimulus identities. Urban uses 3,471 valence trials from 62 participants and 56 images; other scales are not copied into valence/arousal labels. A source recording with confirmed truncation was excluded before prediction inspection, as documented in the frozen coverage manifest. Urban's dataset-description license says CC0, while its README says CC BY 4.0. We document both declarations and do not redistribute the source recording or image material.

SHA-256 ordering with design seed 20260908 assigns participant and stimulus identities to five blocks. Block $b$ supplies test pairs; block $b+1$ modulo five supplies validation pairs; population training initially uses identities from the other blocks. Test pairs lie on diagonal blocks only. Coverage means every eligible participant and stimulus is tested once, not every possible trial. The same test support is retained throughout the dose grid. Stimulus-label injection may use participants from the validation block; only the tested participants are categorically excluded from population fitting at all doses. This is not a claim of fully disjoint validation identities after injection.

## A.2 Bounded model settings and preprocessing

Ridge R1 selects alpha from $\{0.1, 1, 10, 100\}$ and R2 from $\{1, 10, 100, 1000\}$. Both band-power and frozen LaBraM use the two grids. The historical Emo feature key `de` selects channel log-bandpower; it does not make this panel a separate differential-entropy implementation. Standard EEGNet is Braindecode 1.2.0 with F1=8, D=2, F2=16, dropout 0.25, AdamW weight decay $10^{-4}$, batch size 128, at most 100 epochs and patience 15. L1 and L2 use learning rates $10^{-3}$ and $10^{-4}$. Validation MAE selects training length; the selected length is refitted on train plus validation. Both direct and residual branches are retained. Calibration changes post-fit offsets only; it is not a sweep of few-shot fine-tuning algorithms.

Emo uses the existing 100-Hz raw trial archive and ten-second EEGNet crops. Urban uses 64 EEG channels in physical volts, excludes ECG and GSR, applies common-average reference and polyphase resampling from 500 to 100 Hz, and uses the three-second image interval before the rating response. At evaluation, window predictions are averaged within a trial. Downstream learned normalization is fitted without test or calibration trials. Frozen LaBraM uses the project's recorded checkpoint and channel/unit adaptation; its upstream training-data identity provenance is not certified here.

### A.3 Complete primary and secondary summaries

All tables below are generated from the same hashed aggregates as the main tables. No new fitting, bootstrap, or model selection was performed to produce this manuscript revision. The score identity $EC - P = (PC - P) + (EC - PC)$ is checked for all 300 task–setting–dose combinations after block aggregation, with maximum numerical error below $10^{-12}$. This validates a numerical identity, not a causal decomposition. Paired intervals are copied from paired contrast outputs, never constructed by subtracting separate confidence bounds.

Table 5: EmoEEG-MC: uniform-grid EC-PC CCC. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

| Setting | Mean | Percentile interval | t-normal interval |
|---|---|---|---|
| Band-power R1 | -0.0069 | [-0.0333, +0.0221] | [-0.0435, +0.0298] |
| Band-power R2 | -0.0091 | [-0.0306, +0.0145] | [-0.0385, +0.0202] |
| LaBraM R1 | -0.0320 | [-0.0676, +0.0104] | [-0.0842, +0.0201] |
| LaBraM R2 | -0.0137 | [-0.0376, +0.0139] | [-0.0466, +0.0192] |
| EEGNet L1 | -0.0001 | [-0.0040, +0.0047] | [-0.0060, +0.0059] |
| EEGNet L2 | +0.0003 | [-0.0028, +0.0040] | [-0.0042, +0.0048] |

Table 6: EmoEEG-MC: uniform-grid EC-PC MAE. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

| Setting | Mean | Percentile interval | t-normal interval |
|---|---|---|---|
| Band-power R1 | +0.0047 | [-0.0004, +0.0099] | [-0.0019, +0.0113] |
| Band-power R2 | +0.0039 | [0.0000, +0.0078] | [-0.0013, +0.0090] |
| LaBraM R1 | +0.0151 | [+0.0061, +0.0242] | [+0.0034, +0.0269] |
| LaBraM R2 | +0.0065 | [+0.0016, +0.0112] | [+0.0001, +0.0129] |
| EEGNet L1 | +0.0002 | [-0.0008, +0.0013] | [-0.0012, +0.0016] |
| EEGNet L2 | +0.0002 | [-0.0004, +0.0009] | [-0.0007, +0.0011] |

Table 7: Urban: uniform-grid EC-PC CCC. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

| Setting | Mean | Percentile interval | t-normal interval |
|---|---|---|---|
| Band-power R1 | -0.0060 | [-0.0432, +0.0376] | [-0.0519, +0.0398] |
| Band-power R2 | -0.0069 | [-0.0312, +0.0216] | [-0.0370, +0.0233] |
| LaBraM R1 | -0.0173 | [-0.0482, +0.0112] | [-0.0522, +0.0175] |
| LaBraM R2 | -0.0035 | [-0.0238, +0.0150] | [-0.0257, +0.0187] |
| EEGNet L1 | +0.0073 | [-0.0056, +0.0217] | [-0.0080, +0.0226] |
| EEGNet L2 | +0.0086 | [-0.0043, +0.0215] | [-0.0059, +0.0230] |

### A.4 Resource effects and statistical limits

For the EEG-free prior, participant-dose changes from $(0, 0)$ to $(8, 0)$ are $+0.2036$ $[+0.0955, +0.3116]$ on Emo and $+0.1564$ $[+0.0492, +0.2637]$ on Urban. Stimulus-dose changes

Table 8: Urban: uniform-grid EC-PC MAE. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

| Setting | Mean | Percentile interval | t-normal interval |
|---|---|---|---|
| Band-power R1 | +0.0063 | [-0.0012, +0.0144] | [-0.0024, +0.0151] |
| Band-power R2 | +0.0037 | [-0.0009, +0.0082] | [-0.0015, +0.0089] |
| LaBraM R1 | +0.0091 | [+0.0036, +0.0148] | [+0.0027, +0.0156] |
| LaBraM R2 | +0.0038 | [+0.0003, +0.0073] | [-0.0002, +0.0078] |
| EEGNet L1 | +0.0027 | [-0.0009, +0.0068] | [-0.0016, +0.0071] |
| EEGNet L2 | +0.0012 | [-0.0013, +0.0041] | [-0.0019, +0.0043] |

to $(0, 8)$ are $+0.4907$ $[+0.3581, +0.6233]$ and $+0.3408$ $[+0.2161, +0.4655]$. Joint changes to $(8, 8)$ are $+0.6327$ $[+0.4810, +0.7844]$ and $+0.4181$ $[+0.2928, +0.5435]$. These are heuristic t-normal sensitivity intervals; the corresponding percentile intervals also exclude zero. They are not the same contrasts as $PC - P$ at a fixed stimulus dose in the main endpoint table.

The bootstrap reuses crossed participant and stimulus multiplicities across all fitted predictors. Averages are computed within block and then across blocks rather than pretending repeated grid predictions are new observations. The finite-cluster coverage diagnostics and invalid-draw counts constrain interpretation. These are fixed-fit uncertainty summaries, not repeated-training confidence statements or equivalence tests. Neither isolated pointwise intervals nor dose-family bands imply simultaneous coverage over every method, task, and secondary endpoint inspected during benchmark development.

### A.5 Known-truth evaluator stress tests

We tested the estimator on outcomes with known construction before interpreting the EEG results. In the v6 oracle suite, 16 settings with 100 replicates each produced the following operating characteristics. “Detection” denotes a strictly positive interval and is not a universal power guarantee.

| Generator | Recorded behavior | Interpretation boundary |
|---|---|---|
| Labels only | mean difference exactly zero | algebraic negative control only |
| Independent noise | 0/16 cells detected | finite evidence, not a proof of zero false positives |
| Trial signal | detection 0.85–1.00; coverage 0.92–1.00 | detects the injected trial signal in these settings |
| Participant constant | minimum coverage 0.82 | exposes finite-cluster weakness; no biological or causal interpretation |

The v7 finite-cluster study planned 9,600 fits; 9,584 were valid and 16 were rejected by the declared support rule. In the target $32 \times 24$ cluster setting, percentile, basic, and t-normal coverages were 0.933, 0.930, and 0.957. Their minima in the deliberately small $8 \times 6$ setting were 0.837, 0.820, and 0.940. The often wider t-normal interval is therefore retained as a sensitivity analysis, not promoted as a generally calibrated solution.

### A.6 Rule basis and three-state decisions

The checker implements observable evidence rules rather than inferring scientific validity from a score. Every rule returns allow, block, or unverifiable; the last state is preserved when required provenance is absent.

| Rule family | Required evidence | Example boundary |
|---|---|---|
| Trial lineage | trial UIDs in every partition | repeated derived windows block trial-isolation claims |
| Identity holdout | participant/stimulus UIDs | overlap blocks only the identity claim it violates |
| Preprocess/selection | fit and selection UIDs | missing provenance is unverifiable, not a pass |
| Matched resources | predictor-specific label access | unequal calibration blocks an EEG-added-value claim |
| Fixed support | test-set hashes by comparator | unequal test support blocks a paired comparison |

These rules operationalize documented leakage and benchmark-reporting concerns [3, 4, 13, 26]; they do not certify data truth, upstream pretraining history, or causal validity.

# B Source-Linked Execution Cases and Reuse

The executed EEGain commit is `d6892f5586181f345969651a9baedabd828bb1c5`; its `EEGDataloader.split_train_val` is in `eegain/data/loader.py`. LibEER is pinned to `dddff9776dbdae21195fe320dff0a5ba61628a18`; we execute `get_split_index`, `index_to_data`, and the empty-validation branch at lines 100–102 of `LibEER/EEGNet_train.py`. File hashes, selected source ranges, case origin, and outputs are recorded in `external_audit/public_summary.json` and its paired case table. Marker windows are generated from real trial identities, not independent EEG measurements. No external classifier score is inferred from the branch tests.

The same EEGain execution record produces different answers to different contracts: its held-out test participants remain excluded, while an additional requirement that validation trials be disjoint from training can be violated. Changing the claim to what was actually isolated is one valid repair. When a strict validation-trial contract is required, partitioning complete trials before generating windows is the appropriate construction instead. LibEER's explicit train/validation/test path passes the exercised split checks; its empty-validation fallback should not be treated as independent validation. The machine-readable repair record `submission_closure_v11/external_claim_repair.json` verifies this strict-block/scoped-allow pair across all five exercised EEGain folds while holding the pinned function, trial universe, and execution output fixed. The repair changes the supportable claim; it does not alter or re-estimate a model score.

```
python scripts/run_workflow_contract_toy.py toy_contract
python -m openaffect_eeg.workflow_audit \
  --trials toy_contract/trials.tsv \
  --execution toy_contract/unmatched_calibration_execution.json \
  --contract toy_contract/unmatched_calibration_contract.json \
  --output toy_contract/checked.json
```

The mismatch example exits with the documented blocked status; the matched example passes observable checks. Omitting preprocessing evidence preserves uncertainty. This is a developer-operated executable example, not independent human acceptance. The checker validates consistency of supplied execution records, not adversarially falsified provenance.

**AI-assisted development.** AI assistance was used for protocol and code drafting, debugging, literature triage, analysis support, and manuscript development. The recorded EEG predictions, bootstrap summaries, and contract decisions are generated by the documented numerical models and deterministic checks, not by an LLM acting as an emotion predictor or audit judge. Automated checks and agent-run examples are not independent human review. Authors remain responsible for source verification, scientific interpretation, and submission approval.

# C Matched-Grid Display and Learning Diagnostics

## C.1 Complete cell evidence and deployment interpretation

The display in Figure 1 is generated directly from `table_added_value_contrasts.csv`. It preserves all 300 task–setting–dose rows, with their paired pointwise CCC/MAE intervals and within-model dose-family CCC bands. The four resource corners give 48 comparisons across all settings, including unfavorable and uncertain results. The files `matched_increment_all_cells.csv` and `deployment_corners.csv` retain unrounded values. A common symmetric color scale is used, with entries multiplied by 1000 for legibility. The visualization was added after examining the results and is descriptive.

The four corners answer different deployment questions: no target-user calibration or repeated-stimulus ratings; calibration only; repeated-stimulus ratings only; and both. Released normative information remains separately declared. These scenarios do not make ratings costless or assert that stimuli will recur in a real application. A weighted application score requires weights selected for that application and paired prediction-level uncertainty. Marginal interval endpoints cannot be averaged to obtain such an interval.

### C.2 Real-task optimization evidence

Table 9: Previously executed EEGNet training diagnostics, now summarized by task and branch. Counts refer to settings, identity blocks, and stimulus doses, not independent participants or studies. Objective reduction is the median initial-minus-final recorded training objective; these values are not test metrics and are not comparable performance measures across tasks.

| Task | Branch | Fits | Epoch-one | Median epoch | Objective reduction |
|---|---|---|---|---|---|
| Emo | Direct | 50 | 12 | 18 | 0.1104 |
| Emo | Residual | 50 | 10 | 17.5 | 0.1162 |
| Urban | Direct | 50 | 8 | 11 | 0.1922 |
| Urban | Residual | 50 | 7 | 7.5 | 0.2080 |

These aggregates come from the same verification record as the primary predictions, not from newly selected runs. The historical standard/compact EEGNet and matched shrinkage-calibration comparisons remain supporting evidence on their original supports. The repaired real SEED-V task provides a separate compatibility check, not a positive control for the current self-report regression task. None of these records spans full fine-tuning, all calibration algorithms, or repeated-training uncertainty. The scope of the matched increment remains the specified fitted predictor pair.

### C.3 Five-block interval validation

The production statistic first computes paired CCC differences within each identity block, then averages blocks and dose cells. It does not pool all trials before computing CCC. The new validation implementation preserves this order and uses the same global participant and stimulus multinomial weights across all blocks and cells. A draw invalid in any block is not repaired by omitting that block. Tests compare its cell estimates, uniform means, interval endpoints, and valid-draw counts with the production analyzer for both one- and two-target tasks.

The frozen validation configuration uses the exact complete diagonal test supports in the primary verification record: $6\times9, 6\times9, 6\times8, 6\times8, 6\times8$ for Emo, and $13\times12, 13\times11, 12\times11, 12\times11, 12\times11$ for Urban. It retains the full $5 \times 5$ dose grid, two targets for Emo and one for Urban. Four previously defined Gaussian scenarios exercise label-only cancellation, trial signal, participant-constant signal, and independent noise. Each scenario uses 300 fresh simulation replicates per task and 2,000 bootstrap draws. Percentile, basic, and heuristic t-normal intervals are retained as fixed candidates; the production method is not selected using the observed EEG results.

The estimand is the scenario's analytic population matched CCC difference, averaged uniformly over dose cells. Finite-block estimation bias is part of the stress test. Synthetic block resources are independent: matching this test topology does not reproduce all cross-block training-resource dependence, the actual EEG distribution, or retraining uncertainty. Coverage proportions require Monte Carlo uncertainty and do not establish validity outside the tested constructions. Executed results are recorded in the accompanying validation artifact.

Table 10: Five-block uniform-mean coverage, 300 replicates per row and a nominal 95% interval. Brackets are Wilson Monte Carlo intervals for percentile coverage, not EEG effect intervals.

| Task | Scenario | Percentile [MC interval] | Basic | t-normal |
|---|---|---|---|---|
| Emo | Trial signal | 0.900 [0.861, 0.929] | 0.983 | 0.990 |
| Emo | Participant constant | 0.903 [0.865, 0.932] | 0.910 | 0.973 |
| Emo | Independent noise | 1.000 [0.987, 1.000] | 1.000 | 1.000 |
| Emo | Labels only | 1.000 [0.987, 1.000] | 1.000 | 1.000 |
| Urban | Trial signal | 0.927 [0.891, 0.951] | 0.977 | 0.993 |
| Urban | Participant constant | 0.970 [0.944, 0.984] | 0.920 | 0.977 |
| Urban | Independent noise | 1.000 [0.987, 1.000] | 1.000 | 1.000 |
| Urban | Labels only | 1.000 [0.987, 1.000] | 1.000 | 1.000 |

All 2,400 simulation replicates completed without a failed run. The released 187,200 rows include every cell, uniform mean, and candidate interval, not 187,200 independent simulations. Labels-only cancellation gives degenerate zero differences; its coverage of one is an algebraic check, not calibration of a nondegenerate interval. Independent noise produces no positive interval in these runs, which does not establish false-positive control for every null.

The percentile failure is substantive. At Emo's zero-dose participant-constant cell, population-target bias is $-0.0756$ and coverage is $0.193$ (Monte Carlo interval $[0.153, 0.242]$); t-normal coverage is only $0.887$. For trial signal, the worst Emo cell covers at $0.770$. These constructions expose finite-block CCC bias and bootstrap miscalibration. They do not diagnose which component dominates in the real EEG data. The mean-level t-normal coverage of $0.973$–$1.000$ across these scenarios is not a reason to select that method post hoc or claim general calibration. Statistical intervals throughout this revision are conditional diagnostics; the resource identity and descriptive paired differences do not depend on a nominal significance threshold. Calibrated inference would require a separately validated procedure or a redesigned support/estimand and is not claimed complete by this study.

# Historical Supporting Analyses

These analyses retain their original test supports and estimators. They are supporting records, not independent replications of the current primary grid. The current design and diagnostics precede this historical supplement.

# D Data Sources and Trial Reconstruction

## D.1 Version-pinned sources

OpenAffect-EEG uses fixed OpenNeuro snapshots rather than a floating latest version: MusicEEG ds002721 version 1.0.3, DENS ds003751 version 1.0.6, EmoEEG-MC ds005540 version 1.0.7, and external validation ds006866 version 1.0.0 [7, 21, 23, 30]. Snapshot manifests, byte counts or annex keys, and derived trial-table hashes record the source state. Across the 4.0 datasets we reconstruct 39642.0 trials from 279.0 dataset-qualified participants; 38369.0 trials have complete harmonizable continuous targets. The core tier contains 131.0 participants, 4135.0 trials, 369.0 qualified stimulus identities, and 2865.0 supervised trials.

The DENS gap originates from unreadable public signal windows rather than an explicit target filter, but the resulting complete-case set is not ignorable. Only 162.0 of 365.0 label-eligible trials are available (missing fraction 0.5562); 7.0 participants have no available trial, and the available-versus-missing valence standardized mean difference is 0.1895. Missingness summaries and probe seen-class fractions are published per dataset and seed. Subject probes in the LaBraM case study have complete test coverage; permutation labels are shuffled only in the training partition while the assignment, features, estimator, and test labels remain fixed.

**MusicEEG.** Participants listened to film-music excerpts and completed eight self-report items [6]. The benchmark retains 1,240 task trials. Public events do not directly provide canonical audio filenames, so `stimulus_uid` is aligned from event codes to the public OSF Set 1 after checking uniqueness, code range, and the availability of audio and normative ratings for every candidate. Because the README, event description, and early paper do not fully agree on the material set, this mapping is labeled `audited_inference`, not `source_verified`. The OpenNeuro snapshot declares CC0 and the companion OSF resource declares CC BY 4.0; audio is not redistributed.

**DENS.** DENS uses naturalistic videos and collects dimensional and categorical ratings [2]. Events and behavior are joined by normalized stimulus name, never row position. Several public `.fdt` lengths disagree with their `.set` headers; unreadable full windows are recorded through an availability mask rather than padded, wrapped, or treated as valid EEG. Public events contain twenty local stimulus names, but the snapshot lacks the documented stimulus and code directories, and the cited AFDI entry cannot be reliably linked to those local identities. They are therefore valid grouping keys but not claims of recovered source films. Conflicting CC0 and ODbL/DbCL statements are handled under the more conservative attribution and share-alike boundary.

**EmoEEG-MC.** The dataset combines video viewing and text-guided imagery [31]. A published behavioral `video_name` field repeats one erroneous value for some participants and cannot be joined by row order. The supervised subset is restricted to globally verifiable stimulus codes, and context direction is cross-checked against source EDF trigger types. Other reconstructed trials remain in provenance with unavailable targets. Public DE arrays, reordered EEG, model features, and checkpoints are excluded from the anonymous package.

**External emotion regulation.** ds006866 studies social and nonsocial images under neutral viewing, negative viewing, and reappraisal conditions [22]. It contributes 148.0 participants and 35507.0 image-presentation trials, of which 35504.0 have complete valence and arousal. Public events recover condition but not trial-level image identity, and EEGLAB annotations do not restore it. Stimulus and joint holdout are consequently blocked by source metadata. The OpenNeuro snapshot declares CC0; companion metadata declare CC BY 4.0, while source images can have separate terms. Neither images nor a substitute media index are distributed.

Table 11: Source and redistribution boundaries. License entries describe the fixed source metadata; companion media and code retain separate terms.

| Dataset | OpenNeuro snapshot | Companion boundary | Benchmark release |
|---|---|---|---|
| MusicEEG | CC0 | OSF metadata/audio: CC BY 4.0 | Metadata, splits, scores |
| DENS | CC0 field; ODbL/DbCL conflict | Videos not mirrored | Conservative attribution |
| EmoEEG-MC | CC0 | Stimuli/code/checkpoints separate | No EEG, DE, media, weights |
| ds006866 | CC0 | OSF table: CC BY 4.0; source images separate | No EEG or images |

Table 12: Observed-data and probe-coverage disclosure. EEG feature counts are applied after a frozen trial assignment; missing signals never redefine a split. Probe coverage is the test fraction whose class was observed in fit.

| Dataset | Labeled | EEG features | Main role | Stimulus probe |
|---|---|---|---|---|
| MusicEEG | 1,240 | 1,240 | Transparent spectral audit | 0.8708 |
| DENS | 365 | 162 | Transparent, missingness-limited | 0.9826 |
| EmoEEG-MC | 1,260 | 1,260 | DE and foundation case study | 1.0000 |
| ds006866 | 35,504 | 35,504 | Condition-level external audit | N/A |

# E Target Harmonization

MusicEEG pleasantness and energetic-arousal reports define valence and arousal; the other datasets use their released valence and arousal fields. MusicEEG, DENS, and ds006866 map an original one-to-nine scale by $(x-1)/8$; EmoEEG-MC maps zero-to-seven by $x/7$. Original values, scale labels, and transformation expressions remain in trial metadata or result provenance.

The supervised target is the participant's trial-level report. Nominal stimulus category, normative rating, and experimental condition can be inputs, stratification variables, or audit labels, but do not replace the participant target. Trials missing either continuous dimension remain in metadata and are excluded only from metrics requiring both dimensions.

# F Split Protocols

**Single-factor protocols and the optimistic control.** Complete trials are assigned before windowing. Random-trial, subject, and stimulus holdout use 70/10/20 train/validation/test proportions within each dataset, with at least one identity reserved for validation and test in small groups. Random-trial splitting isolates only `trial_uid` and records subject and stimulus overlaps. Strict single-factor splits require an empty train–test intersection for the held identity.

**Joint holdout.** Subjects and stimuli are independently assigned 60/20/20. A trial is retained only when both identities receive the same partition; cross-partition edges are marked `excluded`. This sacrifices coverage in sparse subject–stimulus graphs to obtain a genuine double cold start. The ratio was fixed before model evaluation because smaller validation identity sets did not provide adequate validation coverage across seeds.

**Support-matched identity-overlap control.** For each joint-holdout seed, the control freezes the exact test trial IDs and matches the joint split's train and validation row counts. It then samples those training-side rows so that both test subjects and test stimuli occur in the fit partition. Consequently, the control preserves test support and sample counts while changing training-side identity overlap and the associated graph. It is an estimand for the practical effect of available identity structure, not a claim that all higher-order graph properties are identical. Across seeds, test sets contain 32.0–56.0 trials from 4.0–7.0 participants and exactly 8.0 held-out stimuli.

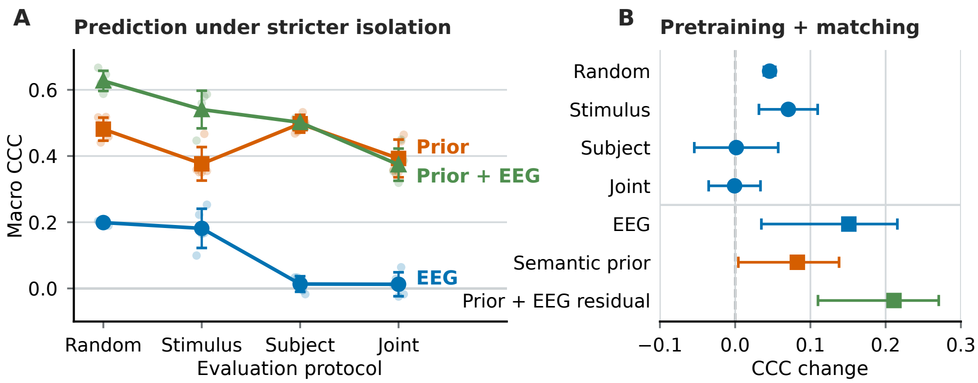


Figure S1: Protocol-level audit retained as supporting evidence. **A:** five fixed split seeds and their mean macro-CCC for semantic prior, direct EEG, and prior-plus-EEG residual; random-trial splitting is an optimistic control. **B:** pretrained versus fixed-random LaBraM comparisons and support-matched identity-overlap effects. These estimates motivate the response-surface analysis but do not by themselves separate participant calibration from repeated-stimulus exposure.

**Context and dataset holdout.** Directional EmoEEG-MC transfer first holds out subjects. Only source-context trials enter training and validation, and only target-context trials from unseen subjects enter test. Video-to-imagery and imagery-to-video are evaluated separately. Context and dataset holdout treat a complete domain as a group.

All main representation audits use 5.0 fixed seeds. Each assignment stores trial roles and an audit JSON. Builders check UID uniqueness, nonempty identity fields, legal roles, and the required empty identity intersections. Models apply target and feature-availability masks only after loading an assignment, so representation missingness cannot silently alter the split.

### F.1 Held-out deployment-selection utility

To test a practical consequence rather than assume that ranking changes help, we compare a validation-cell selector with one global validation-mean selector. Both receive identical predictions from seven candidates: a personalized EEG-free prior and direct/combined predictions from spectral features and frozen pretrained/random LaBraM. Candidates, dose grids, role seed and rules are fixed before this analysis. No candidate is removed based on test performance.

Participant and stimulus identities have disjoint fitting, inner tuning, selector-validation and test roles. All trials containing the other phase's participant or stimulus identities are quarantined, including resource trials. Assignments use constant placeholders for target-block labels. Ridge tuning uses only inner tuning outcomes; selector choices are hashed before test evaluation. Five resource seeds share the same role partition and target support, not five independent cohorts. The EEG-free and EEG predictors receive identical calibration labels and stimulus-label resources.

EmoEEG-MC has 48 trials from six participants and eight stimuli in each of the two phases. MusicEEG has 25 validation trials (six participants, eight stimuli) and 23 test trials (six participants, nine stimuli), after maximum-dose feasibility filtering. The sparse support limits precision and coverage. All 2,000 paired crossed draws are valid. They condition on fits and validation choices, not selector-training uncertainty. Source, assignment and prediction hashes are verified, and a separate centered-moment implementation reproduces the scores and intervals. This is developer verification, not an independent human reproduction.

Neither primary CCC interval excludes zero. Both secondary MAE differences are small and negative, but no practical-importance threshold was established; they cannot replace the primary endpoint or identify EEG added value. The global choices are random-LaBraM combined prediction for EmoEEG-MC and spectral combined prediction for MusicEEG. These choices describe this candidate set and validation block, not population-level model superiority. This analysis uses datasets

Table 13: Held-out deployment-selection comparison. Both policies use the same validation predictions. Differences are conditioned minus global, uniformly averaged across resource cells; intervals are paired crossed 95% intervals conditional on fits and choices. CCC is primary; MAE is secondary and lower is better.

| Dataset | Metric | Global | Conditioned | Difference [interval] |
|---|---|---|---|---|
| EmoEEG-MC | CCC | 0.34221 | 0.34477 | +0.00256 [-0.00113, 0.00601] |
| EmoEEG-MC | MAE | 0.19523 | 0.19464 | -0.00059 [-0.00103, -0.00013] |
| MusicEEG | CCC | 0.24712 | 0.24600 | -0.00113 [-0.01101, 0.01027] |
| MusicEEG | MAE | 0.29405 | 0.29021 | -0.00384 [-0.00675, -0.00094] |

already examined during project development: role isolation protects the current selection procedure, but is not untouched external confirmation.

### F.2 Identity-exposure response surface

Identity availability is not represented as one binary variable. A deployment state can separately specify whether the participant ID is known, whether unlabelled target-participant data are available, how many labelled calibration trials are available, whether calibration comes from the same session, and whether the test stimulus, dataset, and device occurred during fitting. The present response surface varies exactly two labelled resources: $p$, complete calibration trials per target participant, and $s$, other-participant ratings per test stimulus (or experimental condition where stimulus identity is unavailable). Released normative stimulus metadata, where present, is held fixed and reported as a separate prior rather than counted as $s$. It fixes unlabelled target-participant adaptation at zero. Session and device are not separately isolated by the available metadata and are therefore blocked claims, not implicit successes.

For each seed, we begin from one strict participant–stimulus joint assignment. The test set is restricted once to participants and stimuli supporting the predeclared maximum doses; its trial IDs are then hash-fixed for every $(p, s)$ cell. Participant calibration candidates contain a test participant under a non-test stimulus. They never enter population-model fitting. Repeated-stimulus candidates contain a test stimulus rated by a non-test participant. Dose samples are deterministic and nested, so every lower dose is a subset of every higher dose on the same axis.

Adding $s$ repeated-stimulus rows would otherwise increase sample size. Each added row therefore replaces one base-training donor selected deterministically within context and nearest in the two training targets. Population-training size is exactly constant across cells, while validation and test trials are unchanged. Donor selection reads training labels only. The intervention intentionally changes which stimulus identities are represented during fitting; it does not claim all graph or target-distribution properties are causally matched.

Let $f_s$ be a population predictor fit with stimulus exposure dose $s$. For a target participant $i$, labelled calibration produces only the post-fit offset

$$\widehat{\boldsymbol{\delta}}_{i,p} = \frac{1}{p} \sum_{j \in \mathcal{C}_{i,p}} \left( \mathbf{y}_{ij} - f_s(\mathbf{x}_{ij}) \right), \tag{5}$$

with zero offset at $p = 0$. Test prediction is $f_s(\mathbf{x}_{i,t}) + \widehat{\boldsymbol{\delta}}_{i,p}$. Thus a nonzero $p$ cell is explicitly a known-participant, labelled-calibration result; it is not subject independent. The same construction is applied to the prior-plus-EEG predictor and to the EEG-free prior with exactly the same calibration rows. Population and personalized outputs are stored separately.

**Matched model extension and reproducibility.** The EmoEEG-MC extension evaluates EEGNet and frozen EMOD on every existing assignment, asserting identical test IDs, targets, priors, and calibrated priors. EEGNet selects its epoch count by validation MAE, then refits on train plus validation; target-participant calibration does not update network weights. Direct and residual models are fitted separately. Population fits are reused across participant doses only after verifying unchanged fit assignments. The compact model uses the previously specified custom EEGNet-style architecture with zero adversarial loss. All preprocessing statistics are estimated from its fit data.

Table 14: Maximum-resource predictions with resource-matched controls. P: prior; PC: calibrated prior without EEG; E: prior plus population EEG residual; EC: calibrated E. Brackets are paired crossed-identity 95% pointwise intervals. These are not equivalence tests.

| Representation | P | PC | E | EC | EC − PC [95% CI] |
|---|---|---|---|---|---|
| *EmoEEG-MC (p=8, s=8)* | | | | | |
| DE | 0.4809 | 0.5863 | 0.4416 | 0.5599 | -0.0265 [-0.0640, 0.0178] |
| EEGNet | 0.4809 | 0.5863 | 0.4863 | 0.5955 | 0.0091 [-0.0059, 0.0307] |
| LaBraM pretrained | 0.4809 | 0.5863 | 0.4676 | 0.5748 | -0.0115 [-0.0496, 0.0298] |
| LaBraM random | 0.4809 | 0.5863 | 0.4629 | 0.5711 | -0.0152 [-0.0553, 0.0179] |
| CBraMod pretrained | 0.4809 | 0.5863 | 0.4590 | 0.5686 | -0.0177 [-0.0472, 0.0235] |
| CBraMod random | 0.4809 | 0.5863 | 0.4817 | 0.5840 | -0.0023 [-0.0327, 0.0262] |
| EMOD frozen | 0.4809 | 0.5863 | 0.4738 | 0.5819 | -0.0045 [-0.0359, 0.0258] |
| EMOD random | 0.4809 | 0.5863 | 0.4530 | 0.5687 | -0.0176 [-0.0669, 0.0278] |
| *MusicEEG (p=8, s=4)* | | | | | |
| Band power | 0.1740 | 0.2930 | 0.1799 | 0.2988 | 0.0058 [-0.0131, 0.0253] |
| LaBraM pretrained | 0.1740 | 0.2930 | 0.2085 | 0.3070 | 0.0141 [-0.0332, 0.0569] |
| LaBraM random | 0.1740 | 0.2930 | 0.1805 | 0.2941 | 0.0011 [-0.0531, 0.0587] |

EMOD uses official repository revision `3a219f51ba4f90b9e2d22824cfb90dde77a9875c`. Its checkpoint has legacy pooling/projection names; the adapter records a one-to-one name mapping and then requires a strict match of all tensors. It uses the benchmark's 100-Hz raw archive, three nonoverlapping ten-second crops, and mean pooled official projection embeddings. Channels absent from the official channel vocabulary are excluded and listed in the extraction manifest. Input scaling/preprocessing and head adaptation differ from the paper's full fine-tuning pipeline. This audits the supplied frozen weights, not a claim that EMOD's original reported SOTA performance was reproduced. Neither third-party weights nor raw feature arrays are redistributed.

**Failed native-task learning check.** A separate bounded SEED-V check used the official SEEDV model branch and checkpoint backbone with 62 channels, 200 Hz, microvolt inputs and one-second windows. Within each participant/session, the first five, middle five and last five trials formed train, validation and test. Across 48 source files there were 117,744 windows; 40,352 test windows represented 240 trials from 16 participants. A single fixed seed, 100 epochs, and validation-kappa checkpoint selection yielded test balanced accuracy 0.20 and kappa 0: all test windows received one class. Prepared-input hashes, test identities, finite gradients and recomputed metrics were verified, but native-task learning was not recovered. The released SEEDV branch uses a large multilayer head, whereas the paper describes a linear classifier; the per-task selected fine-tuning recipe is not fully specified. This failed compatibility check neither reproduces nor refutes the published scores. It cannot validate the frozen regression audit as a full SOTA test. CUDA pooling backward is nondeterministic despite fixed seeds; this is recorded with the aggregate training history.

**Training-only diagnosis and repair.** We retained that failed run and reproduced its numerical failure on 50 fixed training windows, ten per class, without validation/test-based parameter selection. With classifier learning rate $10^{-3}$, the first update increased the first dense layer's output standard deviation from 0.773 to 32.280; by step 20, the second hidden ELU produced constant outputs. The official SEEDV head flattens 39,680 features into a 2,480-unit layer. Keeping the architecture, inputs, initialization, optimizer and backbone learning rate $5 \times 10^{-4}$ unchanged, reducing only the classifier learning rate to $10^{-5}$ prevented saturation and fitted the diagnostic batch by step 50. This establishes a configuration-specific numerical remedy, not an emotion-mechanism finding. All 48 training-label files matched the official constants. Reprocessing 100 training windows across five source files, including the format-fallback and largest-amplitude cases, matched the prepared float32 inputs exactly; this is selected-file parity, not exhaustive independent data validation.

The repaired run completed the fixed 100-epoch budget. Validation kappa selected epoch 21, after which the test set was evaluated once. Recomputed test BACC was 38.62%, kappa 0.2348 and weighted F1 39.43%; all five classes were predicted. The published five-run BACC is $41.24 \pm 0.86\%$ [5], leaving a 2.62 percentage-point gap to that mean. This single-seed check demonstrates recovered learning, not statistical equivalence or a five-run reproduction. It uses within-participant classification, not the self-report exposure response surface, and cannot establish unseen-user transfer or pretrained benefit without a matched random-initialization comparison. Aggregate traces, input/source hashes and a fail-fast input-sensitivity check accompany the repair. The original failed results remain archived.

**Standard implementation and stronger calibration control.** The added standard regression model uses Braindecode 1.2.0 EEGNet with $F_1 = 8$, $D = 2$, $F_2 = 16$, temporal kernels 64/16, dropout 0.25, and two raw regression outputs. AdamW uses learning rate 0.001 and weight decay 0.0001; validation MAE selects among at most 100 epochs with patience 12. All 25 population states and 125 dose cells use the archived assignments. Its 50 direct/residual fits have finite logged objectives and gradients; selected epochs range from 1 to 24, with 14 fits selecting epoch one. Synthetic learning and training-only tiny-subset fitting checks pass, but do not demonstrate optimal benchmark tuning or held-out emotion information.

An additional empirical calibration control estimates within-participant residual variance $\sigma^2$ and nonnegative noise-corrected between-participant mean variance $\tau^2$ from population-fit data. It multiplies every calibrated offset by $w_p = p\tau^2/(p\tau^2 + \sigma^2)$, with $w_0 = 0$. These are heuristic prediction weights, not identified latent variance components. The same weights apply to $PC$ and $EC$. Table 15 reports uniform cell means; the historical dose-weighted integrals remain separate secondary summaries. Neither calibration rule establishes a positive surface-average EEG increment for the displayed models at the reported precision.

**Revised uncertainty.** The paired tables use 2,000 shared crossed-identity draws across fixed splits. The same draws compare every predictor, dose, and model; a degenerate draw is not repaired by silently dropping a seed. The tables expose valid draw counts. The legacy resource-effect table below preserves the original seedwise bootstrap for traceability and is not the revised estimator used in the main text. In particular, MusicEEG's revised joint interval crosses zero. For ds006866, each fixed split holds out one condition; its condition factor cannot support a within-split resampling claim about new conditions.

MusicEEG's frozen grid is five participant doses by four stimulus doses, ending at $s = 4$. Raising $s$ to eight cannot preserve the same test support in four of the five seeds: their available other-participant observations are insufficient. Only one seed supports that extension without dropping test trials. We do not fill unavailable cells or compare a changed test set as if it were the same response surface.

The artifact emits one deterministic claim card per cell. It records ID and label requirements and blocks latent-emotion disentanglement, causal identity removal, unseen-participant claims when $p > 0$, unseen-stimulus claims when $s > 0$, and session/dataset/device transfer unless those factors are separately held out. This is rule-based compilation from split lineage, not free-form language-model judgment.

## G EEG Representations and Models

**Transparent spectra.** Core transparent features use delta, theta, alpha, beta, and gamma DE or Welch log power. Global views aggregate channels; channel views retain the channel-by-band layout. Standardization is fitted on training data only. Ridge regularization is selected by validation MAE and refitted on train plus validation. Single-band, leave-one-band, single-region, and leave-one-region views test predictive sensitivity, not source localization or causal anatomy.

**CBraMod.** Official pretrained and same-architecture random encoders receive identical EmoEEG-MC trials. Each 30-second, 64-channel, 200-Hz trial is organized into one-second patches; channel and time tokens are mean-pooled into a 200-dimensional trial representation. Source commit, checkpoint SHA-256, trial UIDs, and feature hashes accompany the artifact. The encoder is frozen and only the common downstream head is trained.

**LaBraM.** LaBraM uses the official channel-location map and base checkpoint. Thirty one-second patches are processed as three eight-patch windows and one six-patch window, then pooled with patch-count weighting into a 200-dimensional trial representation. Pretraining masks and decoder heads are excluded, while the official final normalization path is retained. The random control instantiates the same architecture without loading a checkpoint.

Table 15: Implementation and calibration sensitivity. Entries are uniform-grid mean matched EEG added value, $\mathrm{CCC}(EC) - \mathrm{CCC}(PC)$, with 95% paired crossed intervals conditional on fitted predictions. Shrinkage weights are estimated only from population-fit residuals and shared by EEG and non-EEG offsets. These are not maximum-dose scores or equivalence tests.

| Dataset | Representation | Mean offset | Shrunk offset |
|---|---|---|---|
| EmoEEG-MC | Compact EEGNet-style | 0.0089 [-0.0034, 0.0254] | 0.0067 [-0.0064, 0.0253] |
| EmoEEG-MC | Standard EEGNet | 0.0006 [-0.0036, 0.0054] | 0.0006 [-0.0044, 0.0060] |
| EmoEEG-MC | LaBraM | -0.0100 [-0.0392, 0.0279] | -0.0098 [-0.0419, 0.0308] |
| EmoEEG-MC | Frozen EMOD | -0.0001 [-0.0280, 0.0247] | -0.0005 [-0.0302, 0.0263] |
| MusicEEG | Band power | 0.0042 [-0.0094, 0.0188] | 0.0052 [-0.0269, 0.0335] |
| MusicEEG | LaBraM | 0.0182 [-0.0161, 0.0452] | 0.0247 [-0.0176, 0.0568] |

Table 16: Companion error contrasts for the same matched resources. Entries are uniform-grid mean $\mathrm{MAE}(EC) - \mathrm{MAE}(PC)$ with paired crossed 95% intervals. Negative values favor the EEG branch; positive values mean larger error. These marginal intervals do not control error across all representations and calibration variants.

| Dataset | Representation | Mean offset | Shrunk offset |
|---|---|---|---|
| EmoEEG-MC | Compact EEGNet-style | -0.0008 [-0.0030, 0.0011] | -0.0002 [-0.0032, 0.0023] |
| EmoEEG-MC | Standard EEGNet | 0.0004 [-0.0004, 0.0012] | 0.0005 [-0.0005, 0.0015] |
| EmoEEG-MC | LaBraM | 0.0059 [0.0001, 0.0114] | 0.0056 [-0.0009, 0.0122] |
| EmoEEG-MC | Frozen EMOD | 0.0032 [-0.0025, 0.0100] | 0.0029 [-0.0029, 0.0099] |
| MusicEEG | Band power | 0.0013 [-0.0021, 0.0044] | 0.0033 [-0.0045, 0.0108] |
| MusicEEG | LaBraM | -0.0006 [-0.0071, 0.0067] | -0.0010 [-0.0100, 0.0096] |

**Historical compact EEGNet-style control.** The custom task-specific control is trained from EmoEEG-MC waveforms resampled with anti-aliasing to 100 Hz. Each epoch uses one deterministic 10-second training crop; validation and test average three nonoverlapping crops. The network uses temporal convolution, depthwise spatial convolution, separable temporal convolution, and an eight-dimensional latent. It tests architecture dependence, not an exhaustive architecture search.

# H Predictors, Debiasing, and Identity Probes

**Predictors.** `direct_eeg` predicts normalized targets from EEG. `semantic_stimulus_prior` combines one-hot nominal category, context, and source stimulus-intensity code with lower-cased TF–IDF description unigrams/bigrams (at most 1,024 features, sublinear term frequency). A Ridge model is fitted to one mean target row per training stimulus; alpha is selected from $\{0.01, 0.1, 1, 10, 100, 1000\}$ by trial-level validation MAE. The intensity code identifies source stimulus variants and never uses a held-out participant's rating. The historical registry identifier `disentangled_prior_plus_eeg_residual` uses a training-only stimulus report mean for seen stimuli and the semantic estimate for unseen stimuli, forms leave-one-trial-out residual targets, predicts them from EEG, and adds the hybrid prior. All branches share the assignment and validation rule; none reads test targets. The identifier denotes arithmetic residualization and does not claim causal or latent-state disentanglement.

**Debiasing counterfactuals.** We evaluate stimulus centering, conditional covariance penalties, response-preserving stimulus-subspace projection, gradient-reversal residual networks, and adversarial EEGNet. Every method contains a zero-strength setting; adversarial strength, projection rank, and epoch are selected by validation residual MAE only. Test-time residual prediction does not use stimulus ID. A successful counterfactual must jointly improve target performance and reduce the relevant identity effect. Variance shrinkage or inconsistent seed direction is reported as a negative result.

**Identity probes.** Dataset identity is predicted under subject holdout; stimulus identity under subject holdout is evaluated only on classes observed in training; subject identity under stimulus holdout is

Table 17: Legacy fixed-support identity-exposure summary (original seedwise bootstrap, retained for traceability). Values are macro-CCC changes from the zero-resource cell; brackets are crossed participant–stimulus 95% intervals. For ds006866, $S$ denotes experimental-condition exposure because source metadata do not expose trial-level image identity.

| Dataset | Representation | Zero | Maximum | Surface AUC | $\Delta P$ | $\Delta S$ | $\Delta(P{+}S)$ |
|---|---|---|---|---|---|---|---|
| EmoEEG-MC | LaBraM pretrained | -0.0052 | 0.5748 | 0.4604 | 0.1788 [0.0712, 0.2307] | 0.4728 [0.2900, 0.5420] | 0.5800 [0.4072, 0.6219] |
| MusicEEG | LaBraM pretrained | 0.1026 | 0.3070 | 0.2302 | 0.0953 [-0.0097, 0.2037] | 0.1060 [-0.0950, 0.2798] | 0.2045 [0.0157, 0.3320] |
| ds006866 | Band power | – | – | – | 0.2856 [0.2132, 0.3461] | -0.0014 [-0.0071, 0.0042] | 0.3256 [0.2694, 0.3685] |

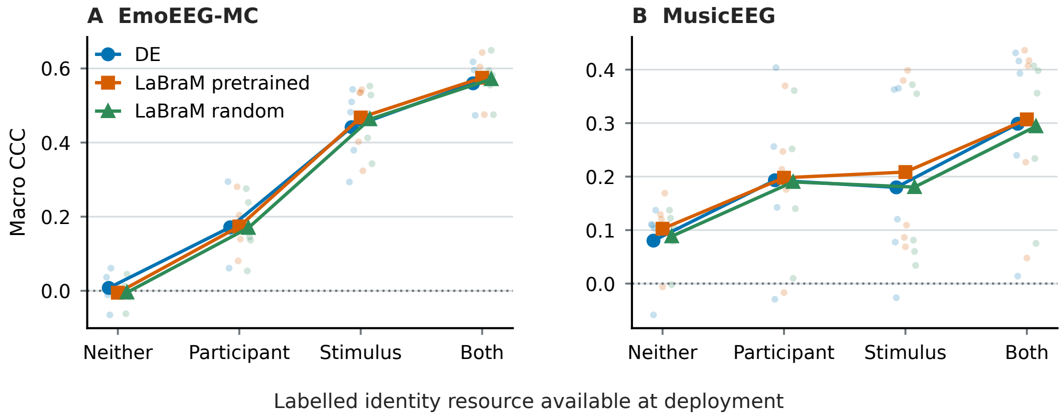


Figure S2: Deployment corners for the three main representations. Large markers and connecting lines show seed means; faint points are the five fixed split seeds. “Participant” uses the maximum labelled calibration dose with unseen stimuli, “Stimulus” uses the maximum repeated-stimulus dose with unseen participants, and “Both” uses both resources. The large MusicEEG spread is why endpoint means are not treated as independent replications.

likewise evaluated on training-seen people. Each probe uses a matched training-label permutation and reports balanced accuracy, class count, uniform chance, coverage, and per-seed values.

**Subject-subspace intervention.** For each stimulus-holdout seed, between-subject centroid differences are estimated on training data, decomposed by SVD, and the leading directions are removed from fit and test LaBraM representations. The requested removal grid is $\{0, 1, 2, 4, 8, 16, 24, 29\}$. At every rank, we also remove the same number of training-estimated PCA directions and seeded random orthonormal directions. Semantic, direct, residual, and subject-probe heads are refitted with the same validation-only selection. The zero-removal point is the nested control. This intervention destroys linearly decodable subject structure but can also remove participant-related affect variation; it is not interpreted as a pure nuisance projector.

# I Statistical Analysis

Continuous predictions report valence, arousal, macro CCC, MAE, and RMSE. Identity tasks report balanced accuracy and, where available, macro-F1. Model comparisons are paired on identical trial-level predictions. In the legacy non-surface analyses, the hierarchical bootstrap first resamples evaluation seeds and then complete held-out identity clusters; joint holdout separately resamples crossed subject and stimulus factors. Those historical intervals use 2,000 resamples and a fixed statistical seed. This is not the revised primary response-surface estimator, which holds split seeds fixed and shares crossed identity weights across every model/cell.

With 5.0 seeds, the exact two-sided sign-flip test has a minimum nonzero $p$-value of 0.0625. We therefore interpret bootstrap intervals as effect uncertainty and the sign-flip test as a finite-seed resolution statement. Historical targeted robustness analyses comprise the support-matched CCC contrast, subject-subspace endpoint CCC contrast, unseen-stimulus LaBraM pretraining contrast, and external condition-prior paired contrasts. These were selected during benchmark development and reviewer-driven audit rather than preregistered. Other reported cells are exploratory. Seeds

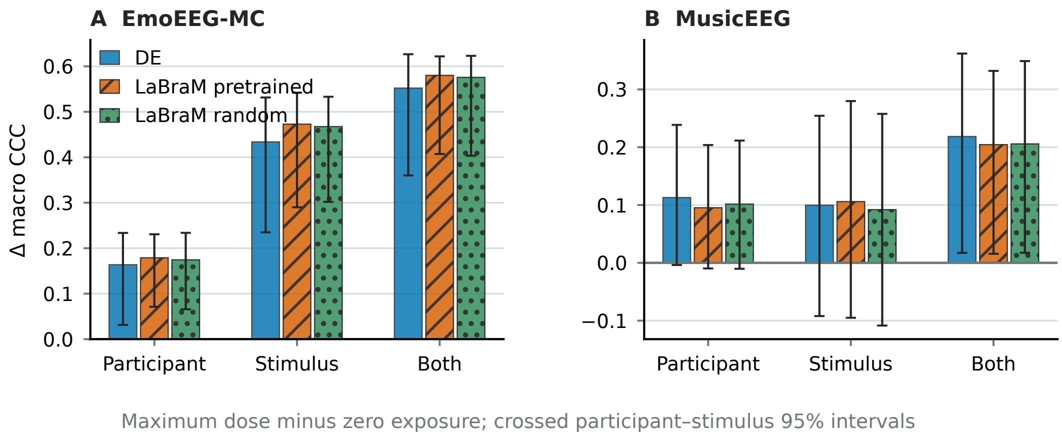


Figure S3: Maximum-dose minus zero-resource changes. Bars distinguish transparent features, pretrained LaBraM, and fixed-random LaBraM; whiskers are crossed participant–stimulus bootstrap 95% intervals. EmoEEG-MC supports positive marginal and joint changes under the legacy estimator. The revised shared-identity interval for MusicEEG's joint change crosses zero; this legacy plot is not evidence of a confirmed external resource effect.

Table 18: Dataset–protocol applicability matrix. "Blocked" denotes missing source identity metadata, not model failure.

| Dataset | Random | Subject | Stimulus | Joint | Context | Dataset |
|---|---|---|---|---|---|---|
| MusicEEG | Yes | Yes | Yes | Yes | N/A | Yes |
| DENS | Yes | Yes | Yes | Yes | N/A | Yes |
| EmoEEG-MC | Yes | Yes | Yes | Yes | Yes | Yes |
| ds006866 | Yes | Yes | Blocked | Blocked | Yes | N/A |

represent Monte Carlo split variability; participant, stimulus, or crossed cluster resampling supplies the corresponding generalization uncertainty. Intervals are pointwise and are not presented as simultaneous family-wise coverage.

Under the same support-matched contrast, the semantic stimulus prior changes by 0.0827 CCC and the hybrid stimulus prior by 0.0991. These branches are reported separately so the full-pipeline effect is not attributed to EEG alone. Component ablations further separate nominal category, context, source intensity code, and description text (Table 23).

The strong nominal-category effect under unseen-stimulus and joint holdout shows that the semantic prior's transferable score is largely experiment-label structure. Description text contributes under subject holdout, whereas context, intensity, and description do not have stable independent joint-holdout effects. These are predictive ablations, not causal interventions on stimulus content.

# J External Validation Details

The initial subject-holdout analysis found that the condition prior exceeded EEG-only macro-CCC by 0.3457, with interval [0.304, 0.3884]. The v2 audit extends this case to exactly matched bandpower, pretrained-LaBraM, and fixed-random-LaBraM representations under five protocols. Table 25 reports every frozen cell rather than only the favorable repeated-participant setting.

For repeated participants, the hybrid improves over the prior in both directions and all five participant-clustered MAE intervals lie below zero for each representation. Under joint participant–context holdout, no LaBraM hybrid interval establishes an improvement over the prior. Pretrained-minus-random hybrid CCC is $-0.0008$ and $-0.0082$ across the two joint directions, with only two of five seeds positive in each. Thus the strict result does not support a pretraining-specific residual advantage.

The near-perfect repeated-participant probes establish information available to a fixed linear readout; they do not show that the affect head causally uses identity. Public events expose condition but not

Table 19: Executed analysis coverage. This table is intentionally distinct from protocol applicability: "supported" does not imply that every model was run in that cell. V/I denotes directional video–imagery transfer.

| Dataset | Representation or analysis | Executed protocols |
|---|---|---|
| EmoEEG-MC | DE, CBraMod, LaBraM affect heads | Random, subject, stimulus, joint, V/I; exposure grid |
| EmoEEG-MC | EEGNet | Joint; complete exposure grid |
| EmoEEG-MC | EMOD frozen and random | Complete exposure grid |
| MusicEEG | Band power, LaBraM pretrained/random | Complete feasible exposure grid |
| MusicEEG, DENS, EmoEEG-MC | Transparent identity probes | Dataset, subject, stimulus as applicable |
| ds006866 | Channel/global bandpower | Subject holdout and condition probe |

Table 20: Representation controls and training state.

| Representation | Trial view | Pretraining control | Downstream state |
|---|---|---|---|
| DE / Welch | Global, channel, band, region | N/A | Validation-selected Ridge |
| CBraMod | Frozen 200-D embedding | Same architecture, random | Frozen encoder |
| LaBraM | Frozen 200-D embedding | Same architecture, random | Frozen encoder |
| EEGNet | Raw deterministic crops | Zero adversary | End-to-end control |
| EMOD | Frozen 128-D embedding | Same architecture, random | Validation-selected Ridge |

trial-level image identity, so stimulus holdout remains blocked. Header-only conformance confirms the same ordered 64-channel EEG set in all 148 recordings; 146 match the base source pattern and two contain one additional, predeclared EMG channel.

## K Authorized SEED-V Protocol-Audit Case Study

SEED-V is a stimulus-induced five-class EEG and eye-movement study with 16 participants, three sessions, and fifteen film clips per session [14]. We report a separately authorized raw-EEG analysis as an independent case study rather than a fifth OpenAffect-EEG source. Its source recordings, feature arrays, media, trial predictions, participant codes, and project paths are not redistributed. The public artifact contains only the source-safe aggregate in `paper/generated/seedv_case_study.json`, its input and code hashes, and this interpretation boundary.

The authorized project compared a raw-temporal baseline against a pre-existing SDSA variant under a strict leave-one-participant-out protocol and a joint participant–stimulus protocol. Table 27 reports all frozen cells rather than selecting the favorable joint cell. The strict SDSA contrast is negative, and both bootstrap intervals include zero. Its frozen decision gate therefore fails; the contrast is evidence of protocol sensitivity, not a model-improvement claim. Separately, the authorized project's strict-to-PaperStyle ladder rises from 0.2786 to 0.5922, so the large protocol gap cannot be attributed to a model variant alone [14].

## L Independent MusicEEG Protocol Audits

To test whether the EmoEEG-MC representation finding is confined to one visual paradigm, we run two prespecified audits on independent MusicEEG [6]. Each uses 1,240 labelled 12-second music trials from 31 participants, five frozen complete-trial splits, and source-safe aggregate reporting. First, a transparent channel log-bandpower Ridge readout reaches macro CCC 0.1031 under random-trial evaluation, 0.0175 with unseen participants, 0.1130 with unseen music but known participants, and 0.0227 under joint participant–music holdout. The paired random-minus-joint difference is 0.0804 (seed SD 0.0357). Thus holding out music alone does not substitute for holding out participants in this auditory source.

Table 21: Optimization and selection budget. Every choice is made on the validation partition and then reused for the paired test comparison.

| Component | Validation search | Training budget |
|---|---|---|
| Ridge prior/EEG/residual heads | Six alphas from $10^{-2}$ to $10^{3}$ | Closed-form per seed/protocol |
| Linear nuisance controls | Five strengths from 0 to 1 | Closed-form per strength |
| Gradient-reversal residual | Six adversary strengths | At most 200 epochs; patience 25 |
| Adversarial EEGNet | Three adversary strengths | At most 100 epochs; patience 12 |
| Subject-subspace audit | Eight ranks, three direction types | Head refit at every point |

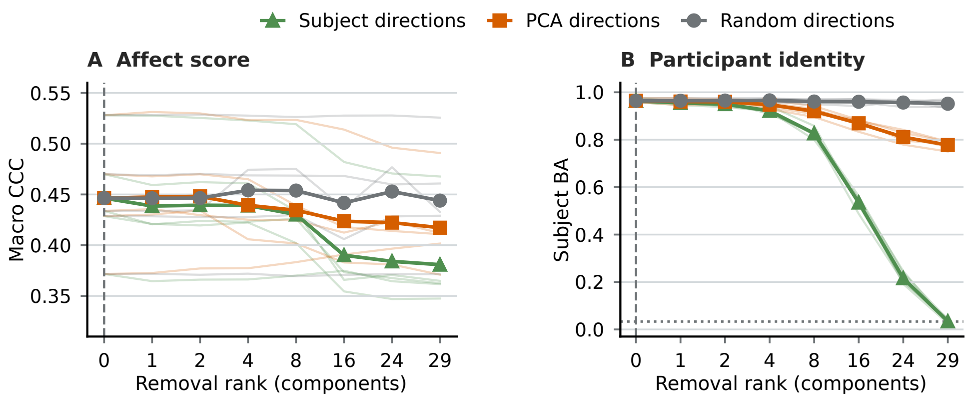


Figure S4: Subject-subspace intervention audit. **A** plots affect macro-CCC and **B** participant balanced accuracy across the fixed removal-rank grid. Faint trajectories are the five evaluation seeds; opaque lines and markers are their means. Triangles remove subject-aligned directions, squares remove equal-rank PCA directions, and circles remove equal-rank random directions. The vertical dashed line is the nested zero-rank control; the horizontal dotted line in **B** is chance. The figure exposes an accuracy–identity trade-off and does not identify participant structure as a pure nuisance factor.

The same bandpower representation identifies training-seen participants under unseen music at balanced accuracy 0.9995 (permuted 0.0431; 31-class uniform chance 0.0323). In contrast, the seen-class music probe under unseen participants is 0.0000 balanced accuracy (permuted 0.0033; 78.8–91.3% test coverage across seeds). These probes establish available information under their stated coverage, not causal use by the affect predictor.

The self-report diagnostic also constrains interpretation. Leave-one-rating-out same-music consensus CCC is 0.1437 for valence and 0.1634 for arousal (1,123 supported ratings each). Descriptive in-sample additive models assign more R-squared to music identity (0.3579 and 0.3484) than participant identity (0.1170 and 0.1782); they are neither test–retest reliability estimates nor causal variance decompositions.

Second, we apply the frozen official LaBraM base encoder to exactly the same MusicEEG split family. Each trial uses its 19 recorded 10–20 channels and its observed event window; MNE volts are converted to microvolts, common-average referenced within trial, resampled to 200 Hz, and final tokens pooled. There is no padding, cross-trial window, fine-tuning, or feature release. Pretrained and fixed-random initializations use the same 200-dimensional Ridge readout, manifests, five split seeds, and nested validation selection.

Table 28 is a separate foundation-model audit rather than a new model claim. Pretraining has a positive matched macro-CCC point difference in all four audited protocols, including participant and joint holdout, but the corresponding unseen-music subject probe is also substantially higher for pretrained features. This external source therefore qualifies, rather than reverses, the main case study: pretrained features have higher report-prediction point estimates in this auditory source while retaining participant structure, so neither the positive gain nor a single identity probe supports a general invariant- emotion conclusion.

Table 22: Selected paired macro-CCC comparisons. Intervals are hierarchical paired-bootstrap 95% intervals.

| Comparison | Protocol | ΔCCC | Interval | Scope |
|---|---|---|---|---|
| LaBraM pretrained − random | Stimulus | 0.0706 | [0.0396, 0.1143] | Known subjects, new stimuli |
| LaBraM pretrained − random | Subject | 0.0015 | Includes zero | New subjects |
| LaBraM pretrained − random | Joint | -0.0008 | Includes zero | New subjects and stimuli |
| Identity overlap − joint, direct EEG | Matched support | 0.1511 | [0.0349, 0.2156] | Same test and fit counts |
| Identity overlap − joint, full pipeline | Matched support | 0.2111 | [0.1101, 0.2706] | Includes prior and residual |
| Max subject removal − zero | Stimulus | -0.0656 | [-0.107, -0.0272] | LaBraM intervention |
| Subject removal − equal-rank PCA | Stimulus | -0.0364 | [-0.0624, -0.0118] | Direction specificity |
| Subject removal − equal-rank random | Stimulus | -0.0631 | [-0.1028, -0.0263] | Direction specificity |
| Condition prior − EEG-only | External subject | 0.3457 | [0.304, 0.3884] | New subjects, repeated conditions |

Table 23: Semantic-prior component audit. Positive values favor the full prior; intervals use the holdout identity required by each protocol.

| Protocol | Omitted component | Full − ablated CCC | 95% interval |
|---|---|---|---|
| Stimulus | Nominal category | 0.3049 | [0.2089, 0.3416] |
| Subject | Description text | 0.0544 | [0.0384, 0.071] |
| Joint | Nominal category | 0.4011 | [0.2557, 0.4591] |
| Joint | Context | 0.0012 | [-0.004, 0.0064] |
| Joint | Intensity code | -0.01 | [-0.0296, 0.0055] |
| Joint | Description text | -0.0129 | [-0.0529, 0.0261] |

## M Compute and Software Environment

Transparent features, linear baselines, probes, paired statistics, artifact compilation, and tests run in a CPU environment on Linux x86-64 with Python 3.11, NumPy 2.4.6, pandas 3.0.5, SciPy 1.17.1, scikit-learn 1.9.0, and MNE 1.12.1. Foundation feature extraction and EEGNet use one NVIDIA RTX 6000 Ada Generation GPU with 49,140 MiB memory, PyTorch 2.11.0, and CUDA 13.0. CPU-only adversarial residual runs use four threads to avoid competing with GPU jobs.

The typed recomputation graph separates feature extraction, evaluation, statistics, and manuscript compilation. Exact historical wall-clock and energy totals were not captured consistently for every preliminary or failed run; the release therefore reports component environments and DAG boundaries rather than inventing a project-wide total. The final cold-run release log is designed to record per-step wall times and will be published with the executable artifact. This limitation does not affect frozen scores but constrains precise compute-cost replication.

## N Release, Licensing, and Reproduction Boundaries

The anonymous package contains code, tests, configurations, evaluation cards, dataset and model cards, Croissant metadata, generated table sources, and claim configuration. It explicitly excludes raw EEG, source feature arrays, foundation checkpoints, movie, music, and image stimuli, office documents, Git history, and machine-local paths.

`metadata` mode validates the benchmark contract without source data. `smoke` uses deterministic synthetic trials to exercise splitting, baselines, probes, and a small registry. `full` requires an explicit user-provided data root and verifies frozen input hashes and paper assets. The typed `recompute`

Table 24: Identity-probe summary. Permuted values use the same representation, split, and estimator with shuffled training labels.

| Representation and task | Holdout | Observed | Permuted | Chance |
|---|---|---|---|---|
| LaBraM pretrained, subject | Stimulus | 0.9642 | 0.0192 | 0.0333 |
| LaBraM random, subject | Stimulus | 0.1733 | 0.0425 | 0.0333 |
| DE, dataset | Subject | 0.8987 | 0.3913 | 0.3333 |
| LaBraM pretrained, stimulus | Subject | 0.0234 | 0.0260 | 0.0238 |
| External channel, condition | Subject | 0.2486 | 0.1676 | 0.1667 |
| External global, condition | Subject | 0.2134 | 0.1692 | 0.1667 |

Table 25: Complete ds006866 v2 affect audit. Entries are five-seed mean macro-CCC. BP denotes 320-dimensional channel bandpower; Pre. and Rand. denote frozen pretrained and fixed-random LaBraM. Hybrid is the transferable condition-family prior plus the EEG residual.

| Protocol | Prior | EEG BP | EEG Pre. | EEG Rand. | Hybrid BP | Hybrid Pre. | Hybrid Rand. |
|---|---|---|---|---|---|---|---|
| Unseen participant | 0.3776 | 0.0369 | 0.0335 | 0.0198 | 0.3425 | 0.3742 | 0.3797 |
| Nonsocial → social, repeated participant | 0.3757 | 0.2319 | 0.2604 | 0.2539 | 0.5144 | 0.5507 | 0.5476 |
| Social → nonsocial, repeated participant | 0.3780 | 0.2226 | 0.2526 | 0.2455 | 0.5116 | 0.5486 | 0.5462 |
| Nonsocial → social, unseen participant | 0.3704 | 0.0208 | 0.0323 | 0.0152 | 0.3318 | 0.3718 | 0.3726 |
| Social → nonsocial, unseen participant | 0.3792 | 0.0172 | 0.0289 | 0.0220 | 0.3472 | 0.3761 | 0.3844 |

`plan/run` interface covers trial construction, feature extraction, evaluation, probes, statistics, and paper compilation; `-resume` skips a step only when all declared outputs exist.

The shortest exact verification sequence from a clean checkout is:

```
conda env create -f environment.yml
conda activate openaffect-eeg
python -m pip install -e ".[smoke]"
python scripts/run_benchmark.py metadata \
  --output-root /tmp/openaffect-metadata
python scripts/run_benchmark.py smoke \
  --output-root /tmp/openaffect-smoke
export D=/path/to/OpenAffect-EEG-data
python scripts/run_benchmark.py full \
  --data-root "$D" --output-root "$D/reproduction"
python scripts/build_neurips_submission.py
```

The first two modes never read EEG. Full mode verifies the frozen registry and rebuilds evidence-linked tables; it is not mislabeled as raw feature extraction or model retraining. The separately documented `recompute_benchmark.py plan/run` commands expose that typed raw-to-paper DAG and require explicit paths to source data, checkpoints, and foundation-model repositories.

Full-mode verification of frozen outputs is not described as a cold download and multi-day retraining run. The existing isolated-environment verification was performed on the same compute host and is not represented as independent laboratory reproduction.

# O Machine-Readable Attachments

The full registry retains per-seed identity results, all paired model comparisons, band and region sensitivity cells, prediction-key audits, source hashes, and unsupported protocol states. These files prevent selective compression of null or unfavorable results into the PDF. Region analyses are predictive sensitivity tests and are not interpreted as neural source localization.

# P Position Relative to Closest Evaluation Resources

Table 26: Matched ds006866 v2 identity probes. Each cell is observed/permuted five-seed mean balanced accuracy. Chance reflects the fitted class set.

| Protocol | Probe | BP | Pretrained | Random | Chance |
|---|---|---|---|---|---|
| Unseen participant | Six-condition | 0.2448/0.1669 | 0.1713/0.1667 | 0.1675/0.1653 | 0.1667 |
| Nonsocial → social, repeated participant | Participant | 0.9910/0.0066 | 0.9997/0.0074 | 0.9963/0.0057 | 0.0068 |
| Social → nonsocial, repeated participant | Participant | 0.9890/0.0063 | 0.9999/0.0045 | 0.9973/0.0046 | 0.0068 |
| Nonsocial → social, unseen participant | Condition family | 0.4091/0.3238 | 0.3399/0.3317 | 0.3306/0.3333 | 0.3333 |
| Social → nonsocial, unseen participant | Condition family | 0.4206/0.3268 | 0.3384/0.3336 | 0.3308/0.3315 | 0.3333 |

Table 27: Authorized SEED-V aggregate audit. Values are mean trial macro-F1 across the frozen result rows; $\Delta$ is SDSA minus raw temporal. No participant-level or trial-level values are released.

| Protocol | Raw temporal | SDSA | $\Delta$ | 95% interval |
|---|---|---|---|---|
| Strict participant holdout | 0.2418 | 0.2297 | -0.0121 | [-0.0439, 0.0178] |
| Joint participant–stimulus holdout | 0.1788 | 0.1856 | 0.0068 | [-0.0154, 0.0292] |

Table 28: Independent MusicEEG frozen LaBraM audit. Values summarize five fixed split seeds; $\Delta$ is pretrained minus fixed-random macro-CCC. Subject probes predict training-seen participants under unseen music.

| Protocol | Pretrained CCC | Random CCC | $\Delta$ CCC |
|---|---|---|---|
| Random trial | 0.0923 | 0.0191 | 0.0732 |
| Unseen participant | 0.0697 | -0.0006 | 0.0703 |
| Unseen music | 0.0847 | 0.0145 | 0.0703 |
| Unseen participant and music | 0.0422 | 0.0085 | 0.0337 |
| Unseen-music subject probe (BA) | 0.9136 | 0.4239 | — |
| Permuted subject probe (BA) | 0.0313 | 0.0297 | chance 0.0323 |

Table 29: The closest resources address complementary layers of the problem. The final column states the additional evaluation object introduced here, rather than asserting that prior work is invalid.

| Resource | Primary evaluation object | Additional OpenAffect-EEG object |
|---|---|---|
| Leakage audit [13] | Trial-segment and test-data leakage on DEAP | Subject, stimulus, joint, context, and dataset identity contracts |
| mdJPT [33] | Multi-dataset affect pretraining and new-dataset transfer | Deployment-specific identity isolation plus matched random encoders |
| EMOD [5] | Discrete/continuous label alignment in a common V–A space | Self-report prior/residual separation and stimulus provenance states |
| EEG-FM-Compass [18] | Cross-task FM comparison under subject transfer and calibration | Affect-specific stimulus and condition audits with predictor-linked interventions |
| FMScope [17] | Subject-axis diagnosis and erasure across clinical EEG-FMs | Joint subject–stimulus–context contracts and report-level affect estimands |
| EEG concept audit [29] | Encoded-versus-used physiological concepts under erasure | Experiment-metadata priors separated from trial-level self-reports |
| Emotion protocol audit [27] | Subject-dependent versus subject-disjoint SEED evaluation | Stimulus, joint, context, and dataset isolation with provenance states |
| OpenAffect-EEG | Executable evidence and applicability contract | Hash registry, claim ledger, cards, blocked cells, and matched controls |

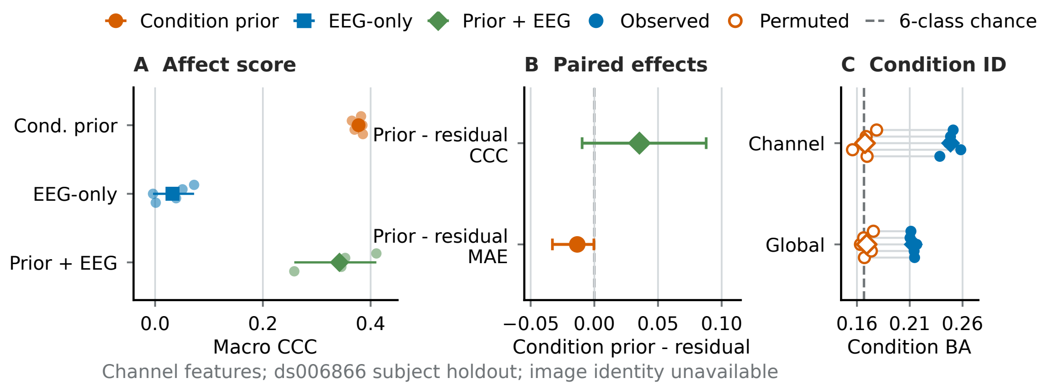


Figure S5: Initial ds006866 subject-holdout condition audit. The condition prior, direct EEG, and condition-plus-EEG residual are reported separately; paired residual effects and condition probes use the five fixed seeds. Image identity is unavailable, so this is condition-level evidence and cannot support a stimulus-holdout claim.

# Q Historical Overview Figures

The following approved figures retain earlier evidence without changing its source version. The initial data layer covers the original four datasets; Urban is the later source described above. These figures are not the latest matched-increment tables and must not be read as latent-emotion decompositions.

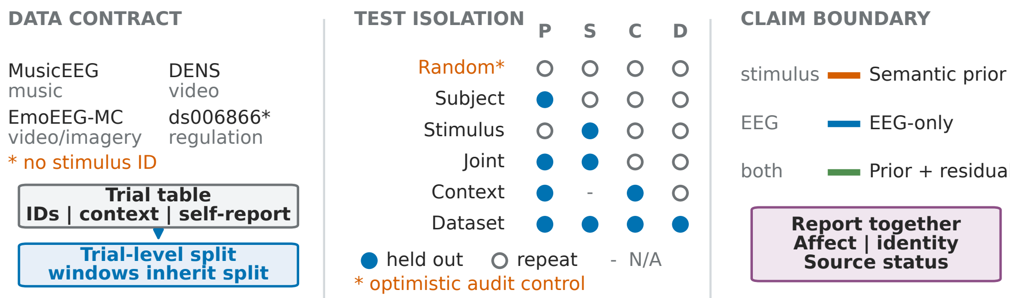


Figure S6: Original trial-first data and evaluation contract. Stimulus identity is unavailable for ds006866. The later Urban source and resource-matched predictor pairs are specified in the main methods and latest-evidence appendix.

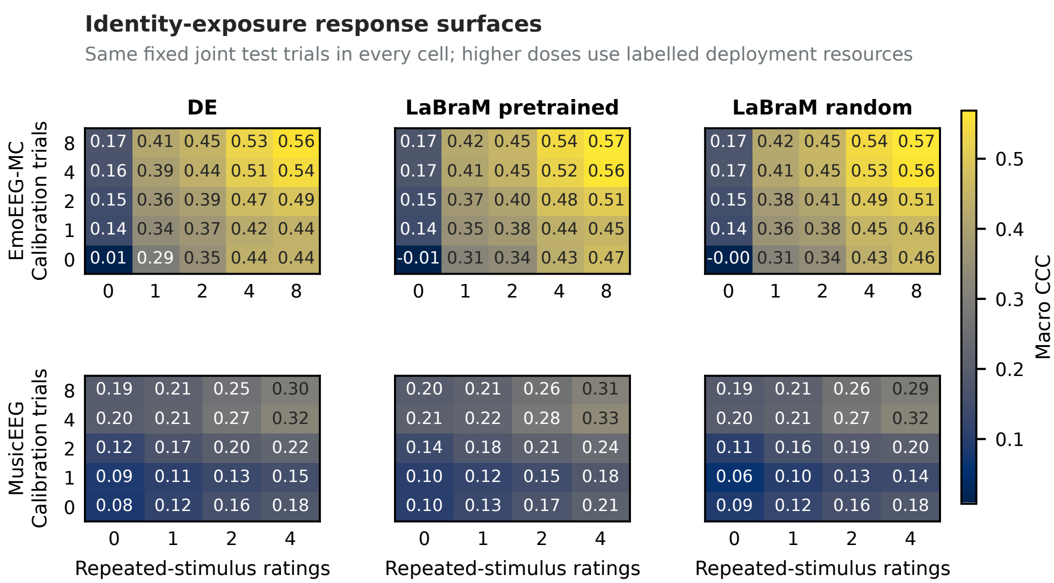


Figure S7: Original EmoEEG-MC and MusicEEG total calibrated prediction surfaces on fixed trials. This is total system CCC, not EEG added value. The new coverage analysis and matched contrasts are reported separately in the main tables; MusicEEG retains its original feasible grid rather than being relabelled as a full maximum-dose grid.

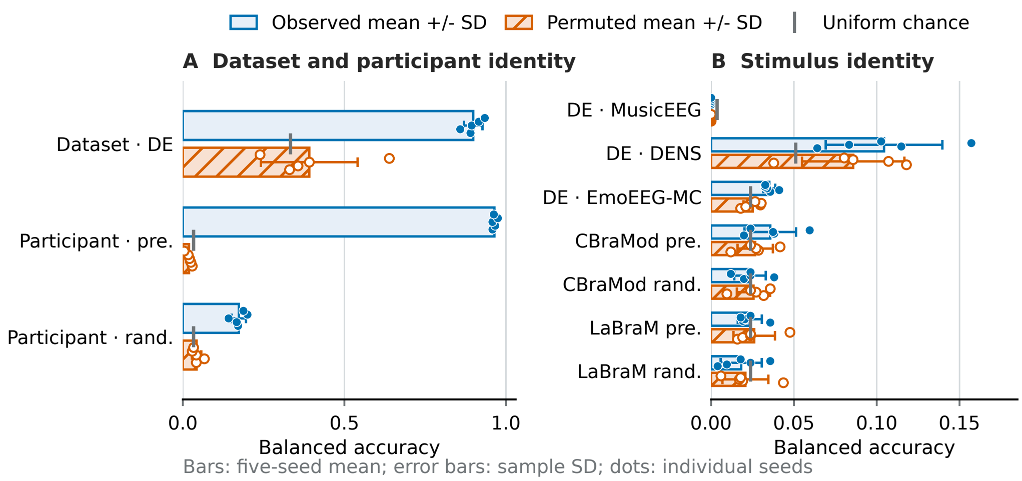


Figure S8: Original representation probes. Bars show the five-seed mean, error bars show sample standard deviation, and points retain every seed. Observed and permuted identity decoding describe available information, not proof that an affect predictor uses identity or that identity is a pure nuisance.

# NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Introduction state the benchmark contributions, identity assumptions, and protocol-dependent scope; the Results and Limitations preserve those same boundaries.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Discussion contains a dedicated Limitations paragraph, with further source-metadata, compute-accounting, and licensing limitations in the appendix.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: Section 3 defines resource-conditioned score contrasts and their arithmetic identity, not a theorem about latent emotion or a causal decomposition.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Sections 3–5, the historical appendices, and the latest-evidence appendix specify sources, trial construction, resource budgets, estimators, paired statistics, and the recomputation commands.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: Code, configurations, aggregates, metadata, and synthetic commands exist in a versioned public release. Unauthenticated access to the anonymous code entry and key files was verified on 9 September 2026. Full raw-to-model reproduction and synchronization of the new revision with that endpoint have not been independently verified, so this working revision retains No. Source recordings remain with their original providers.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 4 gives the evaluation logic and model families; The source, protocol, model, and statistics appendices provide split ratios, adaptation details, validation rules, crops, patches, controls, and statistical settings.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

Justification: The Results and new diagnostics appendix define shared crossed-identity intervals, invalid draws, fixed-fit uncertainty, and heuristic sensitivity; the five-block study exposes serious undercoverage, so no calibrated significance or equivalence claim is made. Historical seed sign-flip tests are separately labelled and are not independent study replication.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

   Justification: The compute appendix reports CPU/GPU type, GPU memory, software versions, and component boundaries, but exact wall-clock and energy totals were not logged consistently for every preliminary and failed run.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work uses consented public research data under source terms, preserves submission anonymity, avoids re-identification, and releases no raw physiological recordings or restricted stimuli.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Sections 6–7 discuss scoped evaluation benefits, privacy and identification risk, and why self-report prediction is not clinical or private-state access.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [Yes]

    Justification: The release excludes raw EEG, source derivatives, restricted media, checkpoints, and real-identity linkage; identity probes are documented as risk measurements rather than identification tools.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: The source and current-evidence appendices cite source versions and unresolved license caveats, including the Urban snapshot's inconsistent declarations. Model cards record code, checkpoint provenance, and licensing boundaries.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The prepared release contains data/model/evaluation cards, Croissant metadata, protocol status, limitations, and reproducibility modes. Anonymous access to key files has been verified; matching the online artifact to this revised manuscript remains a separate version check.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: This work performs secondary analysis of public datasets and recruits no participants or crowd workers; collection procedures remain documented by the cited source studies.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The authors conduct no new human-subject data collection. Ethical approvals and consent for the source recordings are the responsibility of, and documented by, the cited data providers.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: The latest-evidence appendix discloses AI assistance in protocol and code drafting, debugging, literature triage, and manuscript development. EEG predictions and audit decisions are produced by the documented numerical models and deterministic code, not by an LLM. Authors retain responsibility for verification before submission.